Threat Analysis and Risk Assessment in Self-Driving Vehicles

by

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Abstract

With the introduction of self-driving vehicles ensuring automotive safety and security stands as a critical challenge. This research thoroughly investigates the evolving threat landscape to identify new and emerging threats specific to autonomous vehicles (AVs), evaluate limitations and strengths the TARA methodology within the context of AV and identify effective mitigation strategies to safeguard AVs against cyber threats. Through systematic evaluation, the study identifies threats specific to AVs and proposes mitigations for threats in sensor signal interference and manipulation, physical environment manipulation and deception, and unauthorized access and control to make driving decisions. Threats from signal interference and manipulation can be reduced by implementing sensor fusion techniques, advanced cryptographic measures, redundancy and randomization. Physical manipulation and deception can be mitigated by implementing redundancy, physical and operational safeguards, and advanced cryptographic techniques. Unauthorized access and control in automated systems can be countered through advanced cryptographic measures, security protocols, and integrity strategies for software and firmware. The evaluation of the TARA methodology highlights its limitations in applying STRIDE threat modeling and feasibility ratings from experts, particularly when addressing non-traditional cybersecurity threats. These findings contribute to the development of a detailed ontology that categorizes assets, threats, and mitigation strategies specific to AVs, and the consideration of integrating “spoofing” from the STRIDE threat model with “environment” to comprehensively capture AV threats. This study thereby enhances the understanding and management of cybersecurity risks in the automotive industry.

Keywords: Autonomous Vehicles, Self-driving, Threat Analysis and Risk Assessment methodology, TARA, ISO/SAE 21434, STRIDE

Synopsis

Background

The introduction of self-driving vehicles has marked a transformative milestone for transportation systems. The evolving threat landscape extends beyond traditional automotive vulnerabilities, with more ways for remote control and configuration of vehicles. The Threat Analysis and Risk Assessment (TARA) Methodology outlined in the ISO/SAE 21434 standard was developed to address cybersecurity concerns in the automotive industry, aimed at managing cybersecurity risks in automotive systems. Utilizing this methodology, the automotive industry aims to identify cybersecurity risks during the early stages of AV development projects. The effectiveness of TARA could facilitate the identification of evolving threats, enabling the development of mitigation strategies to safeguard AVs against cyber threats.

Problem

The absence of a human driver in AVs creates a new challenge for the automotive industry. Despite the widespread adoption of the TARA methodology, studies highlight its limitations in addressing unique AV characteristics. Further research is needed to identifying emerging threats, evaluating its strengths and limitations in AV contexts, and proposing mitigation strategies.

Research Question

1. What is the evolving threat landscape of autonomous vehicles?
2. What are the limitations and strengths of the TARA methodology under the framework of the ISO/SAE 21434 when applied to address the cybersecurity threats in autonomous vehicles?
3. What are mitigation strategies to address threats in autonomous vehicles?

Method

This study adopts a qualitative approach, employing a case study as the overall research strategy. The data collection method involves document analysis of academic databases, conference proceedings, regulations, and standards. The data analysis method applied is document analysis, systematically examining the documents, extracting relevant information, and identifying key themes in threats and mitigations. Two primary tools, MAXQDA and Microsoft Excel, are utilized. The data analysis method reveals how the TARA methodology translates into actionable steps used to construct an evaluation of the limitations and strengths of the TARA methodology when applied to address cybersecurity threats in AVs.

Result

The study highlighted the evolving threat landscape for AVs, particularly the vulnerabilities of advanced sensor technologies used for navigation and operation. This threat of signal interference and manipulation has significant impacts on AVs. Emerging threats like physical manipulation and deception, and unauthorized access and control, are also identified. To combat signal interference and manipulation, mitigation strategies include sensor fusion, redundancy, randomization and advanced cryptographic techniques. Threats from physical manipulation and deception can be mitigated by implementing redundancy, physical and operational safeguards, and advanced cryptographic techniques. Unauthorized access and control in automated systems can be countered through advanced cryptographic measures, security protocols, and integrity strategies for software and firmware. The evaluation of the TARA methodology revealed its limitations in the use of STRIDE to analyze threats

from environment to AV sensors, and challenges related to expertise in feasibility analysis when dealing with non-traditional cyber security threats.

Discussion

The dynamic nature of cybersecurity threats requires ongoing monitoring and adaptation of strategies, which may not be fully captured in a single study. The findings underscore the vulnerabilities in AV technology, presenting a threat that extends beyond traditional vehicle safety concerns. Without a human driver to intervene or override automated systems determining fault and allocating liability becomes complex. The study proposes two contributions: introducing an ontology that provides an understanding of the unique components in AVs susceptible to cyber threats and their corresponding mitigation strategies, and the consideration of integrating "spoofing" from the STRIDE threat model with "environment" to enrich the TARA methodology in comprehensively capturing AV threats. The study provides a structured analysis of the current threat landscape, evaluates existing methodologies, and suggests enhancements to improve the resilience of AV systems against cyber threats, contributing valuable insights to the field of automotive cybersecurity.

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List of Abbreviations

ABS	Anti-lock braking System
ACN	Automatic Collision Notification
AES	Advanced Encryption Standard
AI	Artificial Intelligence
APE	The Tesla Autopilot Electronic Control Unit
APN	Access Point Name
ASIL	Automotive Safety Integrity Level
ASRG	The Automotive Security Research Group
AUTO-ISAC	The Automotive Information Sharing and Analysis Center
AV	Autonomous Vehicle
CAN	Controller Area Network
CAV	Connected and Autonomous Vehicles
CCAV	Cloud-Assisted Connected and Autonomous Vehicles
CIA	Confidentiality, Integrity, and Availability
CSM	Cab Security Module
CSMS	Cyber Security Management System
DoIP	Diagnostics over IP
DoS	Denial of Service
E/E	Electrical and Electronic
ECUs	Electronic Control Units
EMI	Electromagnetic Interference
ENISA	European Union Agency for Cybersecurity
EVITA	Economic Vulnerability Impact Threat Assessment
FSD	Full Self-Driving
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HARA	Hazard Analysis and Risk Assessment
HEAVENS	HEAling Vulnerabilities to ENhance Software Security and Safety
ISO	International Organisation for Standardization
ITS	Intelligent Traffic Systems
LED	Light Emitting Diode
LiBrA	Lightweight Broadcast Authentication
LiDAR	Light Detection and Ranging
LIN	Local Interconnect Network
MIKEY	Multimedia Internet KEYing
MNO	Mobile Network Operations
MTO	Maintenance, Test, and Operations
NAV	Non-Autonomous Vehicles
NIST	National Institute of Standards and Technology
NMA	Navigation Message Authentication
NTP	Network Time Protocol
OBD	Onboard Diagnostic
OCG	Organised Crime Groups
OEM	Original Equipment Manufacturers
OTA	Over-The-Air
PASTA	Process for Attack Simulation and Threat
RADAR	Radio Detection and Ranging

RF	Radio Frequency
RFI	Radio Frequency Interference
S, F, O, P	Safety, Financial, Operational, and Privacy
SAE	Society of Automotive Engineers
SecOC	Security Onboard Communication
SRTP	Security Real-time Transport Protocol
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege
TARA	Threat Analysis and Risk Assessment
TCU	Telematic Control Unit
TLS	Transport Layer Security
TPMS	Tire Pressure Monitoring System
TVRA	Threat, Vulnerability, and Risk Assessment
UTC	Coordinated Universal Time
V2I	Vehicle-to-Infrastructure
V2N	Vehicle-to-Network
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
VANET	Vehicular Ad-Hoc Networks
VCM	Vehicle Control Module
VPN	Virtual Private Network

1 Introduction

This chapter introduces the security challenges in autonomous vehicles and Threat Analysis and Risk Assessment (TARA) methodology, followed by a description of the problem that this thesis takes on, research aim and question. Lastly, delimitations and thesis structure are presented.

1.1 Background

In recent years, autonomous vehicle (AV) technology has marked a transformative milestone for the automotive industry, promising a fundamental shift in transportation systems and mobility. Self-driving vehicles, equipped with advanced sensing capabilities and AI algorithms, hold the promise of enhanced efficiency and safety in transportation. Pioneering initiatives by companies such as Waymo, Cruise, and Tesla's Full Self-Driving (FSD) have showcased the potential of AV in redefining urban mobility, with robot taxi services and voluntary testing programs showcasing their capabilities across designated areas in the United States and Canada [1].

Chowdhury et al.'s [2] study highlights the importance of numerous connectivity features in AVs. It represents a crucial aspect of development, leveraging various technological advancements to bring closer to the realization of fully autonomous vehicles. These features, alongside intelligent traffic systems (ITS), represent crucial components in advancing towards the realization of fully autonomous vehicles. Leveraging various technological advancements, these connectivity features enable seamless communication and infrastructure support, positioning AVs as highly efficient modes of transport. However, with these advancements come significant challenges, particularly concerning the cybersecurity and resilience of AV systems. The integration of complex connectivity features and millions of lines of code [3] in modern vehicles exposes them to cyber threats, similar to those faced by phones and computers. Without adequate protection, such systems are vulnerable to various cyber-attacks, jeopardizing not only the functionality of the vehicles but also the safety of passengers and pedestrians. Despite efforts to address cybersecurity concerns, the understanding of cybersecurity needs specific to the vehicle is still evolving [4].

One notable development in addressing cybersecurity concerns in the automotive industry is the establishment of the ISO/SAE 21434 standard. The standard initially drafted in February 2020 and later finalized by SAE International and the International Organisation for Standardization (ISO) in September 2021, offers a robust framework for managing cybersecurity risks in automotive systems. Central to the ISO/SAE 21434 standard is the recognition of the Threat Analysis and Risk Assessment (TARA) methodology as a pivotal tool in systematically identifying cyber risks at the early stages of automotive development projects. The methodology outlined in the standard encompasses comprehensive procedures for identifying assets and potential threats, assessing their impacts, and devising effective risk mitigation strategies [5].

As advancements in AV systems progress, ensuring automotive safety and security remains crucial. While the TARA methodology is seen as vital for managing cybersecurity [7], several research presents its limitations when applied to AV-specific assets and properties. These include challenges in addressing unique AV characteristics [8], unclear impact rating criteria, and prioritization issues in cyber-attack assessment [13]. The complexity of attacks targeting digital and physical AV components further complicates traditional analysis techniques [16]. Addressing these shortcomings is essential for enhancing AV cybersecurity. Hence, it is essential to thoroughly investigate the evolving threat landscape to identify new and emerging threats specific to AVs, evaluate limitations and strengths the TARA methodology within the context of AV and identify effective mitigation strategies to safeguard AVs against cyber threats.

1.2 Research Problem

Despite significant advancements in AV technology, ensuring the safety and security of these vehicles becomes paramount. However, the complexity of autonomous systems introduces various threats that could compromise their functionality and pose risks to passengers and pedestrians [9],[10].

The absence of a human driver fundamentally alters the landscape of vehicle security, creating a new set of challenges for the automotive industry [11]. This evolving threat landscape opens ways for adversaries to target AV systems, leaving them vulnerable to both remote and physical attacks [61]. Therefore, adopting a proactive and adaptive approach to cybersecurity is imperative to ensure the sustained safety and security of AV operations amidst constantly evolving threats.

One widely adopted approach is the TARA methodology under the ISO/SAE 21434 framework, providing a structured framework for identifying potential threats, assessing their feasibility and damage impact. Despite its widespread use in automotive manufacturers and suppliers, several studies have highlighted the limitations of the TARA methodology in support of the unique AV characteristics [8]. Challenges such as the traditional threat modeling techniques like STRIDE may struggle to provide comprehensive analyses and effective countermeasures [13]. Additionally, the reliance of the TARA methodology on expert knowledge may introduce biases into the TARA process [8],[14].

Therefore, it is essential to thoroughly investigate the evolving threat landscape to identify new and emerging threats specific to AVs, evaluate limitations and strengths the TARA methodology under the framework of the ISO/SAE 21434 in addressing threats in AVs and identify effective mitigation strategies to safeguard AVs against cyber threats.

1.3 Aim and research question

Research Aims:

- To investigate and analyse the evolving threat landscape of autonomous vehicles.
- To evaluate the limitations and strengths of the TARA methodology under the framework of the ISO/SAE 21434 in addressing cybersecurity threats in autonomous vehicles.
- To identify effective mitigation strategies for addressing cybersecurity threats in autonomous vehicles.

Research Questions:

1. What is the evolving threat landscape of autonomous vehicles?
2. What are the limitations and strengths of the TARA methodology under the framework of the ISO/SAE 21434 when applied to address the cybersecurity threats in autonomous vehicles?
3. What are mitigation strategies to address threats in autonomous vehicles?

1.4 Delimitations of the study

- The study is delimited to the framework provided by the ISO/SAE 21434 standard for automotive cybersecurity.
- The study confines its investigation to cybersecurity challenges and TARA methodology evaluation within the domain of road vehicles categorized under SAE Levels 4 and 5 of automation [17]. These levels denote vehicles capable of high or full automation, where the vehicle can operate without human intervention under certain conditions (Level 4) or under all conditions (Level 5).

1.5 Thesis Structure

Chapter 1 (Introduction) sets the stage for the study by presenting the background on AV cybersecurity, defining the problem statement, introducing the research aims and questions and delimitation. Chapter 2 (Scientific Base) explores cybersecurity principles and the evolution of threats in AV cybersecurity, including mitigation strategies, and the role of standards such as ISO/SAE 21434. This chapter connects the thesis to relevant literature and material in the field. Chapter 3 (Methodology) describes the research approach and methods chosen to investigate the research questions, including a rationale for their selection and a brief discussion on the methods not chosen. Chapter 4 (Application of Data Analysis) provides a comprehensive overview of tools for data collection and data analysis. Chapter 5 (Analysis and Results) reports the research findings, focusing on the evolving threats to AVs and the evaluation and suggested improvements to the TARA methodology, includes mitigation strategy of threat in AVs Chapter 6 (Contribution) offers the development of an ontology for understanding cybersecurity threats faced by AVs and improvement points based on the research findings and insights. Chapter 7 (Conclusion and Discussion) discusses the significance of the findings about the research questions, addresses the research's limitations, and suggests directions for future research in AV cybersecurity.

2 Scientific base

2.1 Introduction to AVs

2.1.1 Definition and Levels of Automation

AV represents a transformative shift in transportation, aiming to replace human drivers with sophisticated computer-controlled systems [18]. Equipped with advanced sensors and actuators, AV is capable of precisely perceiving their environment and executing actions in accordance with established safety standards [19].

The classification is visually represented in Figure 1, which illustrates the progressive stages of vehicle autonomy as recognized by the SAE. into six levels of driving automation [20], ranging from Level 0 with no automation to Level 5 with full automation. Each level signifies a progression in the vehicle's ability to handle driving tasks, with higher levels requiring less human intervention. Notably, semi-autonomous vehicles (Levels 1-3) still rely on human drivers for oversight and intervention, while fully autonomous vehicles (Levels 4-5) operate independently under predefined conditions. The key distinction lies in the reliance on human drivers; semi-autonomous vehicles require a human backup to intervene if the vehicle's system is unable to make a decision based on its surroundings, whereas fully autonomous vehicles operate independently without human intervention [21].

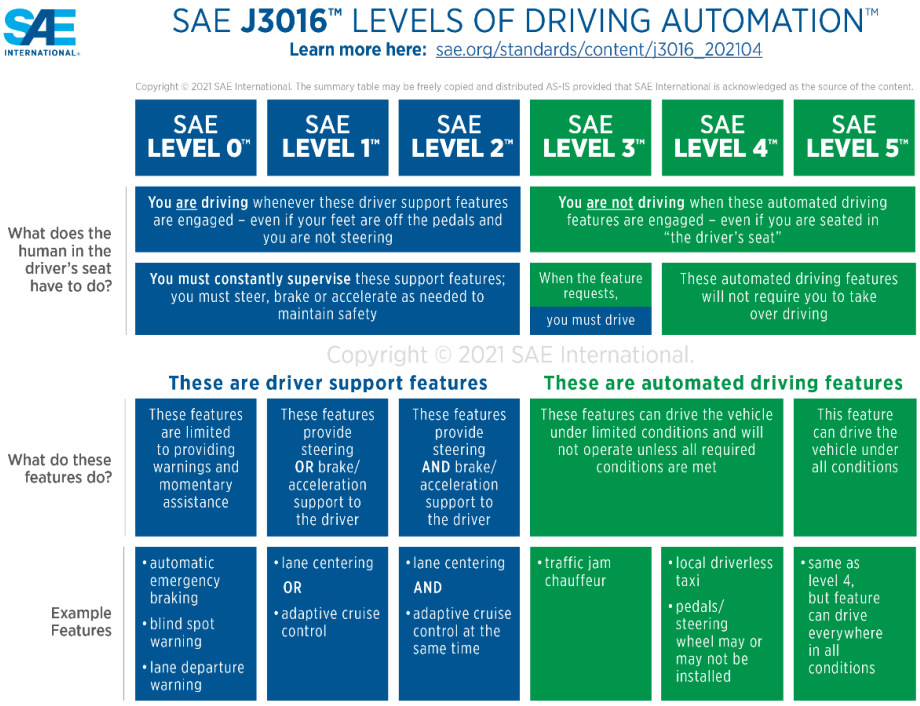

Figure 1. Levels of driving automation by SAE J3016 (2021). Copyright ©2021, SAE International.

2.1.2 AV Technological Foundations

An autonomous vehicle operates as an integrated system, primarily composed of four critical subsystems: a positioning system, a perception system, a planning system, and a control system [22]. Sadaf et al. [23] state that the core technologies driving AVs include a variety of sensors like cameras, Light Detection and Ranging (LiDAR), Radio Detection and Ranging (RADAR), and ultrasonic devices, which help the vehicle perceive its environment by detecting other vehicles, pedestrians, and obstacles. Global Positioning System (GPS) / Global Navigation Satellite System (GNSS) and high-definition maps assist in navigation and obstacle avoidance. Computer vision algorithms analyse sensor

data to identify and track objects, understanding their movement and behaviors. Concurrently, Artificial Intelligence (AI) and Machine Learning are utilized to make decisions, plan routes, predict the behavior of other road users, and respond to unforeseen events.

Furthermore, the control system of AVs relies on an in-vehicle network that connects the main device to auxiliary devices, primarily through Electronic Control Units (ECU). These ECUs play a pivotal role in managing the vehicle's functions, ranging from automatic transmission and engine operation to sensor management. They are divided into body control modules, overseeing features like door operations, seating adjustments, and climate control, and powertrain control modules, which handle the anti-lock braking system (ABS), engine, and transmission management. This network uses technologies like the Controller Area Network (CAN), Local Interconnect Network (LIN), FlexRay, and Ethernet to ensure seamless communication and data exchange between ECUs [24]. Additionally, Vehicle-to-Everything (V2X) communication significantly enhances AV external network interactions, incorporating Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Network (V2N), and Vehicular Ad-Hoc Networks (VANETs). Through V2X, AVs seamlessly integrate into the broader transportation ecosystem, leveraging collective connectivity for improved safety, efficiency, and driving experience. Vehicle ad-hoc networks (VANETs) facilitate dynamic, ad-hoc communication among vehicles and between vehicles and roadside infrastructure, enriching the ecosystem with real-time, decentralized data sharing. Thus, the technologies have been instrumental in enhancing the safety, efficiency, and reliability of AVs [24]. For understanding of how these components interact within AVs, a conceptual diagram illustrating the functional architecture of AVs is provided in Figure 2 [24].

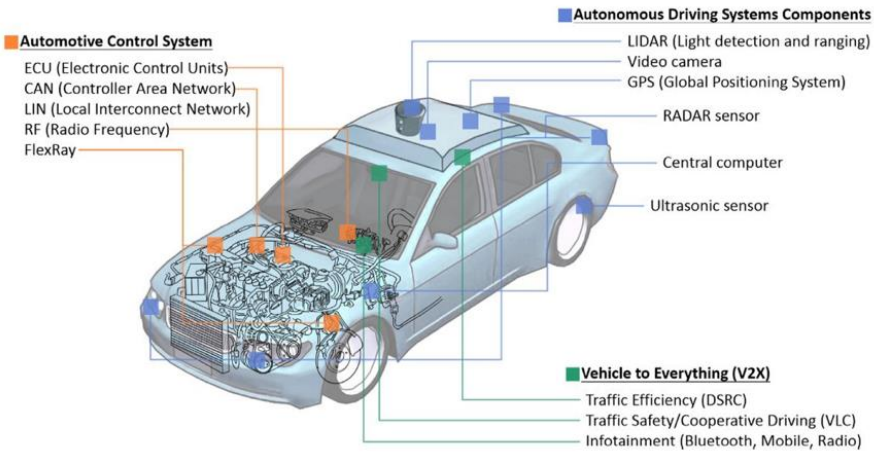

Figure 2. Key element of Autonomous Vehicles

2.1.3 Cybersecurity Principles in the Context of AVs

According to the study by Nobles et al. [19], it points out that the inherent complexity and potential vulnerabilities in their design cause AV to become prime targets for cyber attackers. These vehicles are equipped with 30 to 100 ECUs and integrated computers that form a sophisticated in-vehicle network. Therefore, it is crucial to implement privacy and security measures tailored to AVs to enhance their defence mechanisms against potential cyber threats. Accordingly, the foundational principles of the Confidentiality, Integrity, and Availability (CIA) triad are applied to the security architecture of AV, safeguarding them against a spectrum of cybersecurity risks [23].

2.1.4 Challenges Unique to AVs

Nowadays, automobile companies are embedding advanced technologies in their vehicles, enabling features such as collision detection, accident alerts, speed and traffic signal warnings, alongside location-specific services. This technological integration is driving collaborations between car manufacturers and tech giants, propelling the industry towards AV [25]. The automotive sector is currently navigating a pivotal shift, with Original Equipment Manufacturers (OEMs) moving from traditional designs to fully autonomous platforms. While Tesla has seamlessly incorporated cybersecurity from the outset in its modern platforms, other OEMs are challenged with updating their longstanding vehicle designs, which were not initially conceived for autonomous functionality. The transition to autonomous driving necessitates the replacement of human drivers with advanced perception mechanisms that can interpret the environment with human-like accuracy, requiring specialized security controls within the AV industry [72].

Kukkala et al. [26] highlight that traditional automotive systems prioritize safety and reliability, with security often layered on subsequently, limiting the achievement of comprehensive security goals and increasing vulnerability to cyber-attacks. Despite the growing research in self-driving vehicles, the exploration of cybersecurity challenges remains secondary. In the automotive industry, while safety mechanisms address technical failures, security features protect against malicious attacks, ensuring software behaves as expected even under such conditions. ISO 9126 highlights security as crucial for preventing unauthorized access to software and data. The relationship between safety and security is closely interlinked, especially as vehicle security breaches can directly impact human safety, demonstrating the need for simultaneous consideration of both safety and security to manage threats effectively, such as unauthorized control over critical vehicle functions like engine and brakes [71]. Thus, given the significant risk posed by compromised vehicles, cybersecurity is essential for the seamless operation and safety of AVs. Addressing cybersecurity in the operation and communication networks of self-driving cars is critical [27].

2.2 Introduction to Threats Landscape in AVs

With the integration of sophisticated sensors, AI, extensive vehicular communication systems or even the adoption of over-the-air software updates not only propels vehicles towards autonomy but also significantly expands the attack surface for potential cyber threats [28].

2.2.1 Threats in AVs

Understanding the threats associated with AV is rooted in comprehending the advanced technology and structure of vehicles. The AVs must process data from surroundings, which is gathered through perception sensors, communication with other vehicles, and various intelligent infrastructures [29]. The seamless integration of advanced technologies in AVs enhances the capability for real-time perception and interaction with the environment. However, the integration also exposes AVs to a range of cyber threats that could compromise functionality and safety. A critical vulnerability is the reliance on sensors and cameras for navigation and understanding surroundings. Exploitation of the sensor devices, by feeding false information, can lead to misguided decisions or actions, a tactic known as sensor spoofing and data manipulation. Furthermore, AI and machine learning algorithms that underpin decision-making processes are susceptible to interference. Alteration of the algorithms could change vehicle behavior in potentially hazardous ways [30].

Moreover, the inherent connectivity integrated into AVs marks a distinct departure from traditional, non-AV [31]. Engineered to maintain constant communication, not just internal communication system within the vehicle such as CAN, LIN, FlexRay, and Ethernet communication protocols, but also external connections V2V, V2I, and V2X ecosystem, includes Mobile Network Operations (MNO) and backend operations. The extensive network connectivity expands the attack surface, opening avenues for remote cyber-attacks that could compromise critical systems or intercept personal data [29].

The collective research examines threats to AV, with studies [33] and [36] focusing on threats in V2X communications and in-vehicle networks. Research [14] analyses cyber-attacks on sensors and perception systems crucial for navigation, while studies [21],[24],[30],[32],[34],[35] and [38] assess threats across communication protocols and critical control systems. The body of work highlights the complexity of AV security challenges, stressing the importance of comprehensive safeguards. However, the consensus among many researches are that the enhanced connectivity and autonomous capabilities of modern systems elevate the risk of cyber-attacks. Current research predominantly concentrates on the cybersecurity challenges inherent to these autonomous functions. It is crucial to address potential threats by examining both the autonomous features and the connected aspects of the systems, as highlighted in the study [37].

Thus, this research categorizes threats in AV into three main classes for clearer understanding and mitigation. The first class includes the vehicle's sensors, vital for navigation and perception, such as LiDAR, radar, and camera, GPS/GNSS and IMU which are prone to specific attacks affecting the vehicle's interpretive abilities. The second class encompasses in-vehicle access and control systems, like the immobilizer, keyless entry, and critical control components along with crucial communication networks (CAN, LIN, FlexRay and Ethernet) that facilitate internal component interactions, essential for safeguarding against unauthorized control. The third class covers external network communications, including Maintenance, Test, and Operations (MTO) /Cloud and backend operations, as depicted in Figure 3. The potential threats in AVs are listed and categorized from existing research as shown in Appendix A.

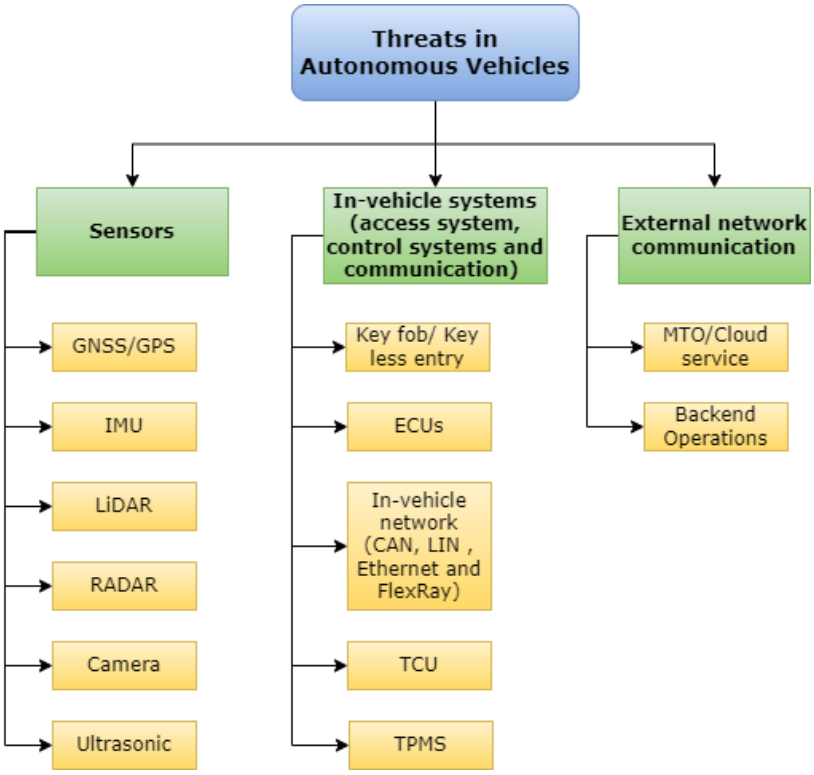

Figure 3. Category of Threats research on AVs

2.2.2 Threat Actors and Motivations

Non-autonomous vehicles (NAVs) are primarily operated by human drivers, making human error the predominant threat actor in this context. Research [62],[63] highlights the role of human factors such as distraction, impairment, and reckless behaviour as leading causes of accidents on the roads. In addition,

NAVs are susceptible to exploitation by criminal actors for illicit activities. Criminal motivations range from financial gain through insurance fraud or vehicle theft to more violent crimes such as vehicular assault or the use of vehicles as weapons in criminal operations. These criminal activities are often driven by a desire for profit or by personal revenges against specific individuals or groups [64],[65],[66].

The emergence of AVs introduces a new frontier of threats characterized by the convergence of cybersecurity vulnerabilities and malicious actors. Numerous studies [39],[40],[41],[42],[43] have identified a diverse array of threat actors as shown in Table 1.

Category	Threat Actor	Example	Motivations
Individual	Outsider	Hacktivist, Script kiddies, Car Owner	Financial Gain, Personal Satisfaction, Passion, Seeking Attention, Ideology, Data Theft, Social Change
	Cyber Criminal	Black Hat	Financial Gain, Data Theft
	Ethical Hacker	White Hat	Financial Gain, Data Theft
	Privileged Insider	Internal Employee	Financial Gain, Discontentment
Group	Established	A group (e.g. the Anonymous group), Organised Crime Groups (OCGs), Cyber Terrorist Groups	Depend on group purpose: Ideological, Financial Gain, Political, Cause major disruptions, Loss of life
Organization	Competitor	An organization about to compete for a tender for services	Corporate espionage; Financial Gain; Reputation Damage
	Supplier	Sub-Contractors, Vendors, Auditors	Information Gain; Financial Gain
	Customer	A customer who feels they have had poor or unfair service	Information Gain; Financial Gain
National	Nation-State	Geopolitical Rival	State Rivalry; Geopolitics

Table 1 Threat Actor and Motivation

Individuals, such as hacktivists, script kiddies, and even car owners, may act as outsiders in this landscape. Car-hacking tools are readily accessible to owners, to bypass manufacturer restrictions [41]. It is foreseen that a growing number of hackers will find the increasing prevalence of vehicles to be highly attractive targets, whether for entertainment or to showcase their cybersecurity expertise, much like they do with computers or cell phones [69]. For instance, they successfully hacked and halted a Jeep Cherokee while it was on a highway, as part of their academic research [69]. Moreover, the recent German researchers [70] discover a secret “Elon Mode” from hacked a Tesla Model 3 by manipulating its inner circuitry, using 520 euros equipment to gain unauthorized access to car data and extract sensitive data from the device’s memory. This included personal information about the driver such as contacts in the phonebook and calendar appointments, but also data about the car such as locations it visited.

Cyber criminals, particularly black hat hackers, prioritize financial gain through data theft and extortion [41]. For example, hacked vehicles can allow stalkers to track victims using GPS/GNSS technology. By offering tracking services for a fee, attackers can profit significantly from this illicit activity [68]. In contrast, White hat hackers leveraging their skills to protect data and systems within legal boundaries [43]. Leenstra et al. [43] underscore that black hat hackers seek financial gain through extortion, malware, and ransomware. Countering them requires disrupting their kill chain. Car manufacturers, such as Tesla adopted responsible disclosure policies, policy enabling white hat hackers to report vulnerabilities for recognition rather than compensation [67].

On the other hand, privileged insiders, including internal employees, may be motivated by financial gain or internal discontent. Some are mercenary insiders, who sell network access, while others are disgruntled employees seeking retaliation due to perceived mistreatment post-employment termination [42].

Established groups, for instance OCGs, and cyber terrorist groups have diverse goals dependent on their purposes. OCGs, equipped with substantial financial and knowledge-based resources, often gravitate towards financial motives, targeting activities such as car theft for profit [41].

Organisational threat actors, including competitors, suppliers, and dissatisfied customers, present unique business risks. Competitors may engage in corporate espionage to gain a competitive advantage,

while suppliers may exploit vulnerabilities in the supply chain for financial gain. Dissatisfied customers might seek information to address unfairness or compensation for perceived poor service, potentially tarnishing the organisation's reputation and impacting its bottom line [43].

Nation-states are pivotal threat actors in cybersecurity, driven by objectives broader than regular cyber criminals, typically aiming to engage in state rivalry and geopolitics while gathering intelligence or advancing national interests [39],[42].

2.2.3 Mitigation Strategies

As AVs continue to evolve, addressing cybersecurity vulnerabilities has become paramount. Mitigation strategies and defences are multifaceted, aiming to fortify AVs against a wide range of cyber threats. The strategies encompass technological solutions and regulatory frameworks to ensure the safety and security of AV operations.

Ren et al. (2020) [32] review existing defence mechanisms and security measures designed to protect AVs from threats surrounding an AV into three categories: sensors equipped in the vehicle, various in-vehicle access and control systems and in-vehicle network protocols. The study encompasses encryption techniques, anomaly detection systems, secure communication protocols, and resilience strategies for AI and machine learning models used in autonomous driving systems. Furthermore, the authors delve into the role of potential practical defence strategies against all the new threats identified, with the goal of offering a comprehensive security guideline.

The study [37] introduces a novel categorization of mitigation methods for attacks on Connected and Autonomous Vehicles (CAVs), highlighting the importance of recovery and protection mechanisms in the cybersecurity of CAVs. This categorization provides valuable guidance for future research, suggesting areas such as intrusion detection and encryption as critical for safeguarding CAV systems. Concurrently, research [34] explores essential aspects for the deployment and advancement of CAVs in the context of cyber-attacks, focusing on the CAV communication system. The study includes both In-vehicle communication and V2X communication, underscoring their significance in combating cyber threats and ensuring the resilience of CAV networks.

The findings highlighted in the study [35] emphasize the critical role of security standards, notably ISO/SAE 21434, in establishing benchmarks for the development of secure systems within the automotive industry. The standard aims to establish uniform minimum security and privacy criteria. Furthermore, the study conducts an in-depth analysis of high-risk threats to Automatic Collision Notification system (ACN) and investigates the root causes and potential impacts. By understanding the threats comprehensively, the study facilitates the development of effective countermeasures, ensuring enhanced protection for ACN systems. Drawing on the mentioned studies, this research offers a comprehensive overview of current mitigation strategies used to protect AVs from cybersecurity threats. Additionally, the European Union Agency for Cybersecurity (ENISA) study [63] focuses on enhancing smart car security through the establishment of critical security measures and best practices. It conducts a thorough analysis of relevant documents and standards to pinpoint key measures for countering threats and risks. The outcomes are organized into three primary areas: policies, organisational practices, and technical practices. Specifically, the technical practices aim at securing smart cars and their back-end systems by addressing software and cloud security, implementing detection mechanisms, and enforcing access control, among other aspects. This effort culminates in a comprehensive list of mitigation measures for each component of AV, detailed in Appendix B, highlighting the study's practical approach to cybersecurity in the automotive sector.

2.2.4 Public Perception and Trust in AVs Security

AVs rely on intricate networks of sensors, software, and communication systems to navigate roads autonomously. Any compromise in these systems could lead to catastrophic consequences, including loss of control [44], accidents [45], or even deliberate misuse for criminal activities [46].

Public perception has been playing a pivotal role in the widespread adoption of AVs [47],[48]. Trust in the safety and reliability of these vehicles is paramount for their acceptance. However, incidents in AVs shatter this trust, instilling fear and skepticism among the public. A survey [47] gathered 5778 responses, assessing public attitudes and knowledge levels regarding AVs. The study found that a 1% increase in knowledge correlated with a 0.65% decrease in interest, a 0.68% decline in trust, a $2466 USD reduction in willingness to pay for AVs, and a 0.56% increase in concerns about traveling in AVs. This shift in attitude is attributed to negative news coverage of AV accidents, which amplifies fear and skepticism about technology. Additionally, when such incidents receive extensive media attention, they significantly impact public perception. The research [48] analysed X (twitter) social media data 15 days prior and 15 days after of high-profile AV crashes of an Uber-pedestrian crash [49] and a Tesla Model X crash [50] in March 2018, finding a 32 % increase in negative tweets about self-driving/autonomous technology, rising from 14% to 46%.

For companies investing in AV technology, customer perceptions are crucial for AV acceptance since customers cannot assess AV technical aspects themselves [51]. A study [51] surveyed 420 consumers to understand what drives intention to use AVs beyond traditional technology acceptance factors by looked at three types of companies: Automotive Incumbents (e.g., Volkswagen, BMW, Ford), Tech Newcomers (e.g., Google, Apple, Baidu), and Mobility Start-ups (e.g., Tesla, BYD).

The study examined how capability reputation, character reputation, and adverse events influence customers. Capability reputation refers to customers' perception of a company's technical expertise in AV development, influenced by innovation and product quality. Character reputation involves customers' view of a company's ethical behavior, affecting trust through transparency and commitment to safety standards. Adverse events such as emissions cheating for automotive firms like Volkswagen, data security issues for technology companies like Google, and fatal accidents for mobility start-ups like Tesla were considered.

The findings indicate that capability reputation has a greater impact than belief in a company's character because technical knowledge and competencies are related to safety issues. Adverse events in automotive incumbents and tech newcomers were perceived similarly severe due to personal impacts, either through car ownership or service usage. Both types of companies, with their long-standing existence, face higher expectations from stakeholders because of their established corporate reputations.

The study highlights that automotive incumbents are viewed as less skilled in developing AVs, with people's interest in using their AVs at a similar level to tech newcomers and only slightly lower than start-ups. This perception may arise from the belief that software and start-up companies excel in future technological development, while traditional manufacturers benefit from their long-standing heritage, which helps mitigate disadvantages in public perception.

2.3 TARA Methodology under ISO/SAE 21434

2.3.1 TARA Methodology

With the rapid advancement of automotive technology, particularly in the realm of software development for vehicles, there's a growing need to address both functional safety and cybersecurity concerns comprehensively [52]. ISO 26262 has long been the established standard for addressing functional safety in automotive systems. It provides guidelines for identifying, mitigating, and managing safety-related hazards throughout the development lifecycle of automotive electronic and electrical systems. However, ISO 26262 does not offer explicit guidance on cybersecurity [53].

In response to this gap, ISO/SAE 21434 was developed to specifically address cybersecurity concerns in the automotive industry [54]. In the context of the standard, cybersecurity refers to electrical and electronic (E/E) systems within road vehicles. The standard provides a framework and baseline for vehicle manufacturers and suppliers to manage cybersecurity risks effectively. Apart from a

standardized cybersecurity framework, the document provides guidance and requirements for assessing and managing cybersecurity risks throughout the lifecycle of these systems.

One essential aspect of ISO/SAE 21434 is the incorporation of TARA methodology, a systematic approach used to identify potential cybersecurity threats and assess the associated risks to automotive systems. The TARA methodology outlined in ISO/SAE 21434 provides a structured approach for conducting threat analysis and risk assessment in the context of automotive cybersecurity [5] (see Appendix C).

It begins with preliminary work, which is item definition of the system or subsystem under consideration along with its environment, functions, interactions, and interfaces. The methodology progresses through various stages, starting with the identification of assets associated with the system and assessing potential vulnerabilities related to cybersecurity properties such as confidentiality, integrity, and availability. Additional properties like non-repudiation, authenticity, and authorization may also be considered [55]. Damage scenarios are identified by evaluating potential compromises to these assets and considering their impact on stakeholders, including factors such as functional safety, financial implications, operational disruptions, and privacy concerns. Threat scenarios are then explored based on identified assets, considering potential attacks using Threat modeling approaches like Economic Vulnerability Impact Threat Assessment (EVITA), Threat, Vulnerability, and Risk Assessment (TVRA), Process for Attack Simulation and Threat Analysis (PASTA), or STRIDE. Potential attack paths within the system are mapped out, and the feasibility of such attacks is assessed based on various parameters. Impact and feasibility ratings are combined to determine risk value. Risk treatment strategies are then decided upon, which may involve avoidance, reduction, sharing, or acceptance of risks. Cybersecurity goals and claims are derived from the assessment results and risk management decisions, guiding the development of cybersecurity concepts [56].

2.3.2 Threat Modeling using STRIDE

Threat modeling using the STRIDE framework, introduced by Microsoft in 2002 [57], is aimed at identifying cybersecurity threats within software systems. This framework classifies threats into six distinct categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.

- **Spoofing:** This threat involves an adversary creating confusion about identities in communication. It is essential to understand that spoofing affects the entity that is deceived, not the one being impersonated. Thus, external elements face a spoofing threat when they are misled about the identity of what or whom they are communicating with.
- **Tampering**: Tampering occurs when an adversary alters data. This alteration can happen as the data travels across a network, while it resides in memory, on disk, or within databases.
- **Repudiation:** In the context of cybersecurity, repudiation involves an adversary denying the occurrence of an action. This can complicate accountability and tracking of malicious activities.
- **Information Disclosure:** This type of threat is concerned with the unauthorized exposure of information. It occurs when sensitive data is revealed to individuals who do not have the proper authorization to view it.
- **Denial of Service (DoS):** A denial of service threat involves an adversary rendering a service inaccessible, thereby denying or degrading service availability to legitimate users.
- **Elevation of Privilege:** This threat involves an adversary obtaining higher access levels or capabilities without proper authorization, often leading to unauthorized access or control over system resources.

Each corresponding to specific security properties such as authenticity, integrity, non-repudiation, confidentiality, availability, and authorization [58]. The versatility of STRIDE enables threat identification throughout various phases of the development life cycle, starting from the design phase of software or hardware, providing valuable insights into potential attack scenarios [59].

In the context of an Electrical and Electronic (E/E) system, the STRIDE model allows for the analysis of each component, with each component being susceptible to one or more threats from each threat category [59]. Table 2 illustrates the association between system components and the corresponding threat categories. The process is exposed to all six threat categories, the external entities are exposed to two threat categories, the data flow is exposed to three threat categories, and the data store is exposed to three threat categories. Furthermore, a single component can face multiple threats within the same category, highlighting the importance of thorough threat analysis and mitigation strategies.

STRIDE Category	External Entity	Process	Data Flow	Data Store
Spoofing	/	/		
Tampering		/	/	/
Repudiation	/	/		
Information Disclosure		/	/	/
Denial of Service		/	/	/
Elevation of Privilege		/		

Table 2 Threat Categories Per Component

2.4 Related Work

Previous research has extensively explored the vulnerabilities of AV systems, focusing on components such as sensors, processors, software, and communication networks [60]. Cyber attackers exploit these vulnerabilities, necessitating ongoing adaptation of cybersecurity measures as AV systems evolve and interconnect [61].

Sheik et al. [61] compiled a comprehensive list of threats to Cloud-Assisted Connected and Autonomous Vehicles (CCAV), emphasizing the significance of dynamic threat analysis in evolving CCAV systems. Protecting CCAVs requires a transition from static defences to dynamic, multifaceted security strategies, highlighting continuous security lifecycle management, using adaptive threat modeling, and Zero Trust principles.

Giannaros et al. [6] conducted a study that delved into both academic literature and real-world incidents to uncover potential threats facing AVs. They specifically focused on vulnerabilities within sensor systems and communication channels, which are fundamental components for the proper functioning of AVs. These vulnerabilities were identified as potential targets for cyber-attacks, which could compromise the safety and reliability of AVs.

In addition to technical challenges, the authors stressed the significance of addressing broader societal, legal, and ethical concerns surrounding AV deployment. They recognized that public trust in the safety and reliability of AVs is essential for their widespread adoption. Achieving this trust necessitates the implementation of thorough safety validation procedures to ensure that AVs meet rigorous safety standards.

Several studies have emphasized on the TARA methodology and its applicability in the domain of AVs. However, challenges and considerations related to TARA were discussed.

Khalid’s study [14] shed light on numerous challenges identified through workshops within the author’s organization, which was tasked with creating a TARA for a new platform project focusing on vehicle safety systems. These challenges include the lack of clarity in the scope of TARAs, emphasizing the need for earlier definition of the level of system analysis. Discrepancies in the understanding and application of assumptions among security team members underscore the necessity for a standardized approach. Moreover, there is a clear demand for precise descriptions of security controls, alongside an analysis of their implementation in residual risk assessments to mitigate the risks of inefficient implementation due to misinterpretation. The heavy reliance on security expert opinion for damage

scenario categories and impact ratings underscores the urgency for enhanced analysis and wider endorsement. A systematic approach is essential for asset identification, while the complexity of attack trees calls for a hierarchical standardized approach to enhance simplicity and reusability.

Costantino et al. [15] provided an opinion on ISO/SAE 21434, viewing it as comprehensive but highlighting the absence of specific intervention thresholds in TARA analyses as it can lead to differing interpretations and actions by companies. This lack of standardized guidelines allows each company to decide how to address identified risks, potentially resulting in insecure practices. For example, different companies may apply varying risk treatments, leading to inconsistencies and potential vulnerabilities across the industry. To remedy this, it is crucial to establish guidelines for intervention thresholds not just for specific items but also for different vehicle domains like chassis and powertrain.

Benyahya et al. [8] conducted a review of TARA methodologies relevant to the automotive sector, highlighting limitations in addressing the specific properties of CAVs of SAE L4 and L5 automation levels. Current methodologies treat risks across different automation levels equivalently, without focusing on unique risks at higher autonomy levels, suggested considering controllability metrics differently when assessing CAVs of SAE L4 and L5. Moreover, the study emphasizes the reliance on expert evaluations for risk estimation, identifying a potential for bias stemming from either overconfidence or underconfidence. In response, the researchers advocate for the incorporation of additional metrics, such as experts' confidence levels, to mitigate potential biases in risk computation.

Kawanishi et al. [16] critiqued the existing TARA methodology in ISO/SAE 21434, identifying issues with the method for impact rating based on SFOP attributes lacks clarity regarding the relative weighting of these attributes and the prioritization of network-based attacks in determining attack feasibility ratings that overlooking significant threats posed by direct access attacks on vehicles' internal systems.

Sheik et al. [13] highlighted the complexity of cyber-attacks on physical components in CCAV, which can exploit vulnerabilities in both digital (software) and physical (hardware) aspects simultaneously. This complexity makes it challenging for traditional approaches like STRIDE and DREAD to provide comprehensive analysis and effective countermeasures. Consequently, there is a need for more robust and comprehensive analysis techniques to ensure CCAV security against emerging threats.

In addition to the previously discussed studies, the United Nations Regulation No. 155 (UN R155) [83] establishes a comprehensive framework for enhancing the cybersecurity of wheeled vehicles, including autonomous vehicles. Adopted under the framework of the 1958 Agreement, UN R155 mandates the establishment of a robust Cyber Security Management System (CSMS) by vehicle manufacturers, focusing on the identification, mitigation, and management of cyber threats throughout the lifecycle of vehicles. The incorporation of UN R155 into cybersecurity strategies is crucial as it not only aligns with ISO/SAE 21434 but also extends the scope of risk assessment and mitigation processes discussed in studies such as those by Costantino et al. [15] and Benyahya et al. [8], by setting specific industry-wide benchmarks for cybersecurity measures.

This research distinguishes itself by not only investigating and analysing evolving threats in AV but also proposing a range of effective mitigation specifically tailored to the unique cybersecurity challenges faced by AVs. To provide empirical validation and practical insights, this research evaluates limitations and strengths of the TARA methodology under the framework of the ISO/SAE 21434, systematically documenting impacts, threats, and risks associated with the AV within the scope of the study. By adopting this hands-on approach, this research aims to contribute not only to theoretical understanding but also to the development of actionable recommendations for enhancing the TARA Methodology when applied to address the cybersecurity threats in AVs.

3 Method

3.1 Research Strategy

The most suitable research strategy for this research is case study, as it allows for conducting an in-depth exploration of the evolving threat landscape and mitigation strategies concerning cybersecurity in AVs. Case study provides an opportunity to delve into real-world scenarios within the natural setting of AVs development and implementation, allowing for a comprehensive understanding of the complexities involved [73].

Moreover, employing a case study approach facilitates a thorough examination of the TARA methodology as a specific instance [73], enabling the assessment of its limitations and strengths when applied to address cybersecurity threats in AVs. By focusing on individual case, this research strategy ensures a detailed analysis of the methodology's effectiveness in mitigating risks within the AVs context.

This research incorporates a case study of the Volvo Group Trucks Technology's implementation of the TARA methodology for cybersecurity in AVs. As a leading automotive manufacturer, Volvo's adoption of the TARA methodology under the ISO/SAE 21434 framework serves as a pertinent example for examination. By examining Volvo's approach, this research can gain practical insights into the real-world application of the TARA methodology and its efficacy in addressing cybersecurity threats specific to autonomous vehicles.

3.1.1 Alternative Research Strategy

Although other research strategies such as experimental research and survey research have their merits, they are not as well-suited to the depth and complexity of the study. Experiments research typically involves controlled settings and manipulations of variables [73], which may not capture the real-world complexities of cybersecurity challenges in AVs. In the case of autonomous vehicles, cybersecurity threats arise from many dynamic and unpredictable factors, including evolving technologies, diverse threat actors, and changing regulatory landscapes. Conducting experiments on these factors may be impractical, unethical, or unsafe. Additionally, the complex interplay between technical vulnerabilities, human factors, and external influences cannot be easily replicated or controlled in experimental settings.

Similarly, survey research may not provide the detailed insights needed to evaluate the TARA methodology in the context of AVs. Surveys typically rely on standardized questionnaires or interviews to gather data from a sample of respondents [73]. However, the intricacies of Volvo's implementation of the TARA methodology and its integration with the ISO/SAE 21434 framework may require a more nuanced and in-depth understanding that surveys may not capture. Additionally, accessing a representative sample of autonomous vehicle manufacturers implementing the TARA methodology might be challenging, as companies may be reluctant to share sensitive cybersecurity information through surveys.

3.2 Case Description

Volvo Group Trucks Technology, a leading automotive manufacturer, is at the forefront of addressing the evolving cybersecurity challenges highlighted in AV technology. The absence of a human driver fundamentally alters vehicle security, introducing a new set of challenges for the automotive industry. By integrating the TARA methodology based on the ISO/SAE 21434 standard, Volvo exemplifies its commitment to tackling the unique threats that AV systems face, including vulnerabilities to both remote and physical attacks. Their involvement in the HEAling Vulnerabilities to ENhance Software Security and Safety (HEAVENS) project in 2016 set a new standard in automotive cybersecurity. This

pioneering collaboration between academia and industry specialists revolutionized threat analysis and risk assessment integration, aligning directly with this research aims to investigate and analyze the evolving threat landscape of AVs.

This case study illustrates Volvo's proactive and adaptive approach to cybersecurity for ensuring the ongoing safety and security of AV operations amidst constantly evolving threats. The analysis showcases how Volvo addresses the limitations and strengths of the TARA methodology in the context of AVs, responding to the challenges where traditional threat modeling techniques may struggle. The HEAVENS project not only influenced Volvo's strategic application of TARA but also underscored the organization's efforts to identify and implement effective mitigation strategies against cyber threats in AVs. This commitment reinforces Volvo's reputation for safety and innovation while directly relating to the research questions regarding the effectiveness of cybersecurity measures in AVs.

3.3 Data Collection Method

Given the research questions to delineate and evaluate the myriad threats, mitigation strategies, and countermeasure technologies relevant to AVs, documentary research emerges as the most fitting approach. This method's preference for comprehensive coverage, along with its capacity to tap into both historical and cutting-edge knowledge, clearly aligns with the research question's demands. Among the varieties of documentary research, this study prioritizes electronic sources accessible via online academic databases. The preference is substantiated by the electronic sources' timely availability, expansive reach, and the dynamic nature of cybersecurity and AV technology, which mandates access to the most current research outputs. The selection also encompasses regulations and standards, alongside the integration of open-source web resources. This inclusion is motivated by their unparalleled ability to supply up-to-date, real-world data on cybersecurity threats and their countermeasures.

The conclusion of this documentary research is a detailed synthesis of the collected data, presented with precision and depth in the research's findings in the Appendix. This variety surpasses traditional methods like face-to-face interviews or physical archives in relevance, accessibility, and breadth of coverage for the study's thematic focus. Denscombe [73] highlights the efficacy of this research method, especially its cost-effective and efficient access to a wide range of data. This approach aligns well with the chosen methodology, leveraging open-source resources for up-to-date, pertinent cybersecurity information, thereby underlining the unmatched efficiency of this method in gathering essential, focused data.

3.3.1 Alternative Data Collection Method

An alternative approach, such as conducting interviews, can be considered. Interviews offer a valuable means to explore complex subjects, including opinions, emotions, and personal experiences, especially when delving into topics requiring in-depth discussion or access to privileged information. While interviews can gather straightforward data, their true value lies in capturing the intricacies of human experiences and perspectives [73]. However, the feasibility of accessing interviewees, as well as the time and costs involved, should be carefully weighed to ensure practicality given the researcher's resources and the geographical spread of participants.

Simultaneously, the use of questionnaires as a data collection method is not deemed suitable for this research which aims to delve deeply into the evolving landscape of threats and the strategies for mitigating cybersecurity risks in AVs. This investigation also includes an in-depth examination of the TARA methodology as a particular case study. While questionnaires can be effective for collecting quantitative data and general qualitative insights, the nature of this research demands a more nuanced qualitative analysis. Such analysis is better suited to unpacking complex scenarios, delving into technical specifics, and gathering expert opinions [73].

3.4 Data Analysis Method

The research primarily collects textual data from academic papers, reports, standards documents, and other electronic sources. Document analysis is selected as the chosen method for analyzing and interpreting these documents. This approach enables a comprehensive examination of various text-containing documents, facilitating qualitative analysis, extracting relevant information, identifying key components, and achieving a thorough understanding of their content [74].

Document analysis is particularly suited for identifying potential cybersecurity threats and vulnerabilities specific to AVs. By scrutinizing the content of previous studies, including academic papers and reports, this research can uncover common attack vectors, security weaknesses, and emerging threats that pose risks to AVs. The method also enables the discovery of mitigation techniques by synthesizing insights from diverse sources such as academic papers and reports to extract relevant insights and strategies proposed by experts in the field. Such alignment with the goal of document analysis helps to elicit meaning, gain understanding, and develop empirical knowledge from textual documents [74]. The process of data extraction and interpretation also organizes information into major themes.

Moreover, document analysis facilitates a thorough review of the TARA methodology by systematically examining the document. This research can discern the specific guidelines, procedures, and recommendations embedded within the document. The process unveils how the TARA methodology translates into actionable steps and guidelines for industry practitioners which can be utilized to construct an evaluation of the limitations and strengths of the TARA methodology when applied to address cybersecurity threats in AVs.

3.4.1 Alternative Analysis Method

Another method to consider for analyzing textual data is content analysis [73], which involves systematically categorizing and quantifying content to identify patterns, themes, or trends. However, it may not be suitable for this research due to its focus on quantification and categorization, which may overlook nuanced qualitative insights. Content analysis prioritizes word frequencies and surface-level characteristics, potentially missing deeper interpretive insights, and the contextual richness of textual data. In contrast, document analysis allows for a more holistic and qualitative examination, enabling researchers to extract relevant information, identify key components, and understand underlying meanings and implications.

3.5 Research Ethics

In selecting documents for analysis, the research adheres to ethical considerations regarding authenticity, credibility, representativeness, and meaning, as outlined by Flick [86]. The research relies on academic papers, reports, and standards documents, including those from reputable organizations such as ISO, National Institute of Standards and Technology (NIST), MITRE, and ENISA. These documents serve as foundational sources of knowledge and best practices within their respective fields.

Academic papers often undergo rigorous peer review processes, ensuring a high level of credibility and reliability. Reports published by esteemed organizations like NIST and ENISA are based on thorough research and expertise, making them valuable resources for understanding industry standards and emerging trends. Similarly, standards documents issued by ISO and other regulatory bodies provide guidelines and benchmarks that are widely recognized and adopted across various sectors.

These documents are carefully assessed for compliance with stringent criteria, including factors such as the source's reputation, the rigor of the research methodologies employed, and the level of expertise demonstrated in the content. By meticulously evaluating these factors, the research ensures that its analyses are based on trustworthy and representative documentation.

For proprietary documents, such as Volvo's TARA methodology, the research exercises utmost discretion and confidentiality. This practice involves refraining from disclosing proprietary information and anonymizing any sensitive data pertaining to individuals or organizations involved. By adhering to these practices, the research upholds privacy rights and maintains the integrity of the information being analyzed.

4 Application of Methods

4.1 Application of Data Collection

The research employs documentary research, focusing on AVs cybersecurity. This strategy is comprehensive, tapping into electronic sources for historical and contemporary insights. The methodological foundation is built upon the guidelines provided by Denscombe [73]. Careful planning is essential in qualitative research to ensure the collection of meaningful data from a variety of credible sources. Below is a step-by-step guide on how to effectively carry out the data collection process.

1) Defining the scope

Before diving into data collection, it is crucial to define the scope of the research clearly and guide the search for documents and ensure that the research remains focused on its objectives. This step involves identifying key areas of interest such as

- Types of threats to AVs.
- Current mitigation strategies.
- Emerging countermeasure technologies.

2) Identifying relevant databases and sources

This phase is foundational, as it directly influences the scope, depth, and quality of the information gathered. To ensure a robust and thorough investigation, a diverse array of resources is leveraged, encompassing academic databases, conference proceedings, regulations and standards, and open-source web resources. The following outline is a deeper dive into the methodology behind selecting and utilizing these sources.

- **Academic Databases:** Utilizing platforms such as IEEE Xplore, Google Scholar, arXiv, and the ACM Digital Library, the research taps into a vast reservoir of peer-reviewed articles, papers, and research findings. The selection process involves using specific search terms and filters to retrieve documents that are most relevant to the study's objectives, ensuring a comprehensive coverage of AV cybersecurity.
- **Conference Proceedings:** Sourcing from specific academic conference websites allows access to the forefront of research and developments presented by field experts. This source involves identifying and targeting conferences that specialize in cybersecurity and autonomous vehicles, ensuring the collection of cutting-edge insights and data.
- **Regulations and Standards:** Essential for understanding the legal and best practice framework surrounding AV cybersecurity, documents such as ISO/SAE 21434, UN regulation No. 155 (UN R155), and specific industry instructions like the TARA from Volvo are meticulously reviewed. This instruction includes examining the latest editions and amendments to stay abreast of current standards.
- **Open-source Web Resources:** Engaging with platforms like MITRE ATT&CK, CVE, and industry groups such as The Automotive Security Research Group (ASRG) and the Automotive Information Sharing and Analysis Center (AUTO-ISAC) provides access to up-to-date threat data and community-driven insights. This source involves regular monitoring and participation in these platforms to gather real-time information and trends.

The validity of the information from these sources is critically assessed based on four core criteria: authenticity, representativeness, meaning, and credibility [73]. This rigorous evaluation ensures the genuineness, accuracy, and relevance of the documentary evidence, safeguarding the integrity and reliability of the research findings. Through this detailed and methodical approach, the research lays a solid foundation for acquiring a comprehensive understanding of the current state and challenges in AV cybersecurity.

3) Process of Data collection

This phase involves a meticulous approach to selecting, analyzing, and synthesizing information from a variety of sources to build a comprehensive understanding of the subject. By employing a strategic method for gathering data, the researchers ensure that the work is both current and relevant to the field's ongoing developments and challenges. The following outlines detail the approach taken in this crucial phase.

- **Key words selection:** When searching through selected databases and sources, use specific keywords related to AV cybersecurity threats, mitigation strategies, and technologies. These keywords are carefully chosen to encompass the extensive range of thesis topics. They include precise terms and Boolean operators such as "autonomous vehicles", "connected vehicles", "cybersecurity", "V2X communication", "autonomous operation", "threat analysis", "self-driving technology", and "in-vehicle network security", ensuring comprehensive coverage and relevance.
- **Filtering Search Results:** The search results are filtered based on their relevance, date of publication and the extent of each document's contribution to the research scope. This approach ensures the gathering of current and significant literature, highlighting the latest developments and ongoing challenges within the field. Specifically, documents chosen for exploring the TARA methodology in ISO/SAE 21434 are selected from August 2021 forward, coinciding with the standard's official publication.
- **Data Extraction:** For each selected document, extract key information aligned with the research objectives, including authors' findings, methodologies, results, and conclusions.

4) Organizing collected data

In this research, Microsoft Word and Excel are collaboratively utilized to record and organize collected data efficiently, offering a comprehensive approach to managing information pertinent to the research objectives. Standardized Microsoft Word templates are crafted for uniform data recording, featuring sections for document titles, sources, key findings, and assessments of authenticity, representativeness, meaning, and credibility. Parallelly, a Microsoft Excel spreadsheet is set up for tracking literature reviews. The spreadsheet includes columns for researcher names, research titles, reference details, sources, methodologies or results or notes on relevance and comments. This systematizes the review process and enables quick sorting and filtering of literature based on various criteria, aligning closely with the research objectives.

4.2 Application of Data Analysis

For this research, document analysis serves as the chosen method for analyzing the extensive textual data collected, with two primary tools utilized: MAXQDA and Microsoft Excel.

4.2.1 Application of Data Analysis: MAXQDA

MAXQDA provides a comprehensive suite of tools specifically designed for qualitative data analysis, making it well-suited for analyzing textual documents in-depth [75] The application facilitated the following steps.

1) Data Organization

The first step involves importing the textual data collected from various sources into the MAXQDA software. This includes academic papers, reports, standards documents, regulations, and any other relevant electronic sources that provide insights into the evolving threat landscape and mitigation strategies. A hierarchical structure is created to categorize documents based on their relevance to different aspects of assets in AVs.

2) Annotating and Coding

In the process of annotating and coding data, researchers systematically categorize segments of text based on their content. This involves creating codes related to threats and mitigations in different aspects of assets components in AVs. Researchers annotate relevant text passages by applying appropriate codes, systematically categorizing them to accurately label each segment according to its content and relevance.

3) Summarizing Key Findings

Researchers summarize key findings extracted from documents using features for creating summaries and generating reports, The software's visualization is aid in interpreting, summarizing key findings by visual representations in thematic maps and coding hierarchies.

4.2.2 Application of Data Analysis: Microsoft Excel

The research utilized Microsoft Excel to provide a structured environment for organizing data from the document analysis of the ISO/SAE 21434 standard document. Each step of the TARA process was allocated to specific spreadsheets, enabling a systematic and comprehensive evaluation of the methodology in addressing cybersecurity threats in AVs.

1) Item Definition

The Item Definition is created to provide essential details about the items under evaluation, including their description, operational environment, boundaries, and interfaces. It provides a structured template for documenting the purpose, functionality, and components of the item, alongside describing the conditions and scenarios in which it operates. Additionally, the item's boundaries and interfaces with other systems, both internal and external to the vehicle. Relevant information about the operational environment relevant to cybersecurity, including assumptions and known vulnerabilities, is also documented. Table 3 exemplifies an item definition of VCM [12],[84],[85], showcasing how this information is organized and presented.

Item Definition
Vehicle Control Module (VCM) is an electronic control unit in a vehicle that manages various functions related to the vehicle's operation. The VCM receives information from sensors throughout the vehicle, processes this data, and then uses it to control different systems, such as the engine, transmission, braking, and steering systems, ensuring they work together smoothly and efficiently. VCM components include Central processing unit (CPU) for executing control algorithms, memory for storing firmware and temporary operational data.
Operational Environment: The VCM operates within the electronic and mechanical ecosystem of the vehicle, requiring robustness against physical vibrations, temperature extremes, and electromagnetic interference. Additionally, it exists within a networked environment for updates and data transmission.
Interfaces: 1)Internal, VCM interfaces with many vehicle systems, including the engine control unit (ECU), transmission control unit (TCU), braking system, steering system, and various sensors and actuators throughout the vehicle. It processes inputs from these systems to manage vehicle performance and safety features. The communication interfaces for internal - CAN, Ethernet. 2)External, VCM connects to diagnostic equipment, the manufacturer's servers for updates and telemetry, and potentially to consumer devices such as smartphones via Bluetooth or Wi-Fi for vehicle status monitoring.
Operational Environment Relevant to Cybersecurity: Assume that all external communication channels have encryption and secure authentication methods. Assume reliance on a secure public key infrastructure (PKI) for cryptographic operations, with the assumption that certificate authorities are properly managed and secure.

Table 3 Example of item definition for VCM

2) System Model

The System Model serves as a foundational resource for comprehending the architecture and intricacies of the AV system under analysis in this research. The system model offers a visual representation of

the AV system's structure, delineating its various components, subsystems, interfaces, and operational functionalities.

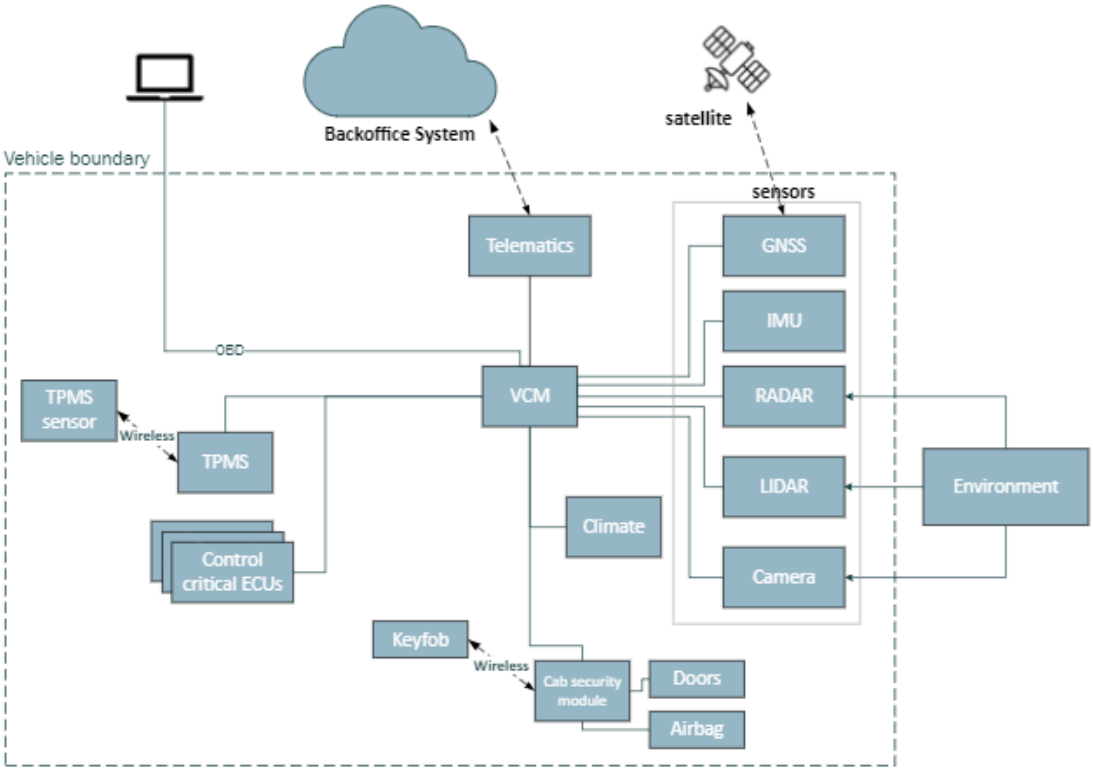

Figure 4. Simplified Architecture of AV System Model

This research utilizes a reference architecture, developed in collaboration with Volvo, as a foundational framework. It is important to note that this simplified reference architecture has been created specifically for the context of this thesis (see Figure 4). The actual architecture of Volvo's autonomous vehicles is notably more intricate and comprehensive.

The AV architecture, as represented here, encompasses numerous interconnected components and subsystems. These elements function collaboratively to ensure the safe and efficient operation of the autonomous vehicle includes both in-vehicle boundaries and exterior boundaries, providing a holistic view of the system's functionality and interactions.

2.1 In-Vehicle Boundary

- **Vehicle Control Module (VCM):** The VCM serves as the central control unit for managing critical ECUs within the vehicle. It interfaces with various sensors, actuators, and subsystems to control vehicle functions such as propulsion, braking, and steering.
- **Telematics:** The telematic serves as the communication bridge, enabling data exchange and commands between the AV and the Backoffice system. It facilitates wireless connectivity, allowing real-time communication for remote monitoring, tracking, and control of the vehicle.
- **TPMS (Tire Pressure Monitoring System):** The TPMS monitors tire pressure and temperature. It consists of sensors installed in each tire to monitor tire pressure and temperature. Each TPMS sensor wirelessly transmits data to a central TPMS control unit or module within the vehicle then data transmits to the VCM for real-time monitoring and alerts.
- **Control Critical ECUs:** The Control Critical ECUs are responsible for managing essential functions within the vehicle, such as propulsion, braking, steering, stability control, and powertrain performance. These ECUs are interconnected and communicate with each other and with the VCM to coordinate vehicle operation and ensure safe and efficient performance.

- **Climate ECU:** The climate control unit manages the vehicle's heating, ventilation, and air conditioning (HVAC) system, ensuring passenger comfort and safety.
- **Sensors:** The vehicle is equipped with a variety of sensors to perceive its environment and detect potential hazards.
    - GNSS: Provides accurate positioning and navigation data.
    - IMU: Measures vehicle acceleration, orientation, and angular velocity.
    - LIDAR: Uses laser pulses to generate 3D maps of the vehicle's surroundings.
    - RADAR: Uses radio waves to detect objects and obstacles in the vehicle's path.
    - Camera: Captures visual data for object recognition, lane detection, and traffic sign recognition.
- **Cab Security Module (CSM):** The Cab Security Module is responsible for managing security functions of the key fob, doors, and airbag systems. It interfaces with these components to ensure secure operation and protect against unauthorized access.
- **Controller Area Network (CAN) protocol:** It is intricately designed to facilitate real-time, low-latency communication within a vehicle's network. It serves as the foundational network, enabling crucial exchanges between various critical ECUs that manage essential vehicle functions. CAN ID allows for precise targeting and coordination among devices, ensuring flexible yet accurate communication across the vehicles critical control systems. The CAN ID is key to the efficient and reliable operation of in-vehicle networks, supporting the broadcast of command data to and from important ECUs within the vehicle.
- **Ethernet:** The technology ensures high bandwidth and low latency communication, essential for real-time decision-making in AVs. It facilitates reliable, high-speed data exchange among critical components such as LiDAR, RADAR, cameras, processors, and control units. This connectivity is crucial for tasks like environment perception, decision-making, and vehicle control. Ethernet also enhances vehicle network architecture by supporting robust, scalable designs and standard IP-based services. This improvement facilitates smoother integration with external networks and cloud services, crucial as vehicles increasingly connect to the IoT ecosystem.

2.2 Exterior boundary

- **Backoffice System:** The Backoffice System serves as a central hub for managing and coordinating vehicle operations remotely. It handles tasks such as route planning, fleet management, and data analysis. It communicates with the vehicle through the telematics system, enabling wireless connectivity for transmitting commands, receiving status updates, and accessing vehicle data.
- **Service Laptop:** The service laptop provides technicians with a means to perform diagnostics, maintenance, and software updates on the vehicle. It connects to the vehicle's onboard diagnostic (OBD) port.
- **Satellite:** The satellite provides additional connectivity and positioning capabilities, enhancing the vehicle's navigation and communication capabilities.
- **Environment:** The environment component represents the external surroundings and conditions in which the AV operates. Factors such as external interference, electromagnetic interference (EMI), radio frequency interference (RFI), or other sources of signal disruption may affect the vehicle's sensors, communication systems, or navigation equipment.

3) Damage Analysis

The Damage Analysis facilitates the damage that could occur in various scenarios, considering road user impact factors which are operational, safety, privacy, financial. Additional impact categories may also be considered. Each damage scenario is categorized based on impact severity, classified as severe, major, moderate, or negligible. Safety-related impact ratings are derived from ISO 26262-3:2018 [76],[77].

Table 4 is an illustrative example of a damage analysis, showcasing key elements which are the damage scenario and road user impact rating.

Damage Analysis	Assumption	Road User Impact Factors			
		1. Operational	2. Safety	3. Privacy	4. Financial
Vehicle theft	Assume the driver have insurance cover theft of the vehicle	Severe	Negligible	Negligible	Moderate

Table 4 Example of damage analysis for the "Vehicle Theft" damage scenario

4) Risk Assessment

The Risk Assessment is used to identify threat scenarios and the targeted assets using STRIDE threat modeling approaches.

Threat Scenario

The development of a Threat Scenario involves a comprehensive analysis that extends from the Cybersecurity Asset to the Threat Category. To ensure the accurate capture of these critical components, a specific syntax is utilized during Threat Scenario Identification. This structured approach aids in not only identifying the threat agent but also in discerning their motivation, when possible. Below is a straightforward guide designed to assist in the description of threat scenarios:

Threat Scenario Syntax
An Attacker launches a *STRIDE - Threat type attack* + on *Cybersecurity Asset*
Example: An attacker launches DoS attack on the TPMS sensor software.

In the specific context of a spoofing threat, it is crucial to understand both the syntax and the underlying meaning. Essentially, the focus of concern is the entity that is being impersonated. For example, if an attacker (malicious process) is pretending to be Process A in order to deceive Process B into believing it is communicating with Process A, then the spoofing threat is directed at Process B. This means that, in the realm of threat modeling and mitigation, Process A is identified as the primary target of the spoofing threat.

For example, the threat scenario described was: "*An attacker spoofs the key fob to the Cab Security Module software*." In this scenario, the implication is clear:

- Process A (the entity being impersonated) would be the key fob in this scenario. It is the legitimate device that the Cab Security Module expects to communicate with.
- Process B (the entity being deceived) is the Cab Security Module software. It is being tricked into thinking it is interacting with the real key fob.
- The attacker (the impersonator) is the entity pretending to be the key fob (Process A) to deceive the Cab Security Module (Process B).

After outlining the threat scenarios, this research established a link between each identified threat scenario within every Cybersecurity Asset and the corresponding damage scenarios. It is important to note that a single threat may correspond to multiple damage scenarios. Table 5 provides an illustrative example of risk assessment for the CSM asset, showcasing the structured approach employed in evaluating and addressing cybersecurity risks within the AV system.

This research will not delve into the risk treatment decision process as it pertains to business decisions. While risk treatment decisions often involve strategic business considerations such as financial implications, resource allocation, and organizational priorities, this study focuses primarily on the technical aspects of risk assessment and mitigation within the context of cybersecurity threats in AVs.

Damage Scenario	Asset	Threat Category	Threat scenario	Risk Level
Vehicle theft	CSM	Spoofing	An attacker spoofs the key fob to the Cab Security Module software.	Critical/High/Medium/Low/None

Table 5 Example of risk assessment for the CSM asset

5) Attack Analysis

Attack Analysis is an analysis of threat scenarios and potential attack paths. This research utilized Attack Tree, employing a top-down approach to deduce attack paths by analyzing the different ways in which a threat scenario could be realized as shown in Figure 5. Each attack scenario is described with sufficient detail to assess their feasibility levels, employing logical operators "and" and "or" to delineate sequential or parallel attack steps as shown in Figure 6.

Furthermore, attack feasibility ratings are assigned to each attack scenario to assess the feasibility of successful execution by adversaries. The methodology for determining these ratings can be tailored to suit specific organizational needs, with options including attack potential-based, CVSS-based, or attack vector-based approaches.

In addition to Microsoft Excel, Visio software was utilized to visually construct attack trees, complementing Excel's analytical capabilities. Attack trees offer a structured representation of potential attack paths. Visio's features enhanced the clarity of these representations, facilitating a comprehensive evaluation of the TARA methodology [78].

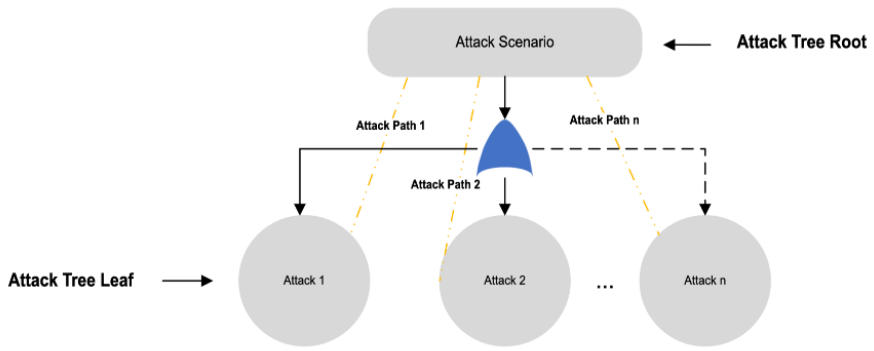

Figure 5. Attack tree scheme showing how a threat scenario is decompose into multiple attack paths

Figure 6. Logical decomposition gates used in attack tree structure

5 Results

5.1 Evolving threat landscape of AVs

In this section, to answer the research question "*what is the evolving threat landscape of autonomous vehicles?*" The result is scrutinized through comprehensive document analysis and conducted via the Threat Analysis and Risk Assessment (TARA).

5.1.1 Threats identified through documents

The threats identified in academic databases, conference proceedings, regulations and standards, and open-source web resources provide a broader perspective on cybersecurity risks. The data has been successfully structured into Table 1 (see Appendix A), providing a clear view of each entry categorized by asset name, threat, threat scenario, consequences, and references. With this structured approach, all data were imported to MAXQDA software for coding and analysis to identify themes and patterns within the data.

Themes were identified based on the type of threat, the assets it affects, and the consequences. For instance, threats like "Jamming" and "Spoofing" appear across different assets, indicating a recurring theme of signal interference. Similarly, consequences related to "vehicle access" and "unauthorized actions" suggest themes around unauthorized access and control. While patterns analysed the frequency of threats by category, the assets most affected, and the most common types of consequences. This reveals insights such as which assets are most vulnerable to specific threats and the potential impact on vehicle functionality and safety. The findings revealed that LiDAR is the primary target for attackers, followed closely by GPS/GNSS devices and ECUs. This result suggests that adversaries may prioritize disrupting vehicles' sensing capabilities and control systems. Among the threats, jamming emerged as the most common, highlighting signal interference as a widespread challenge affecting various vehicular systems. Moreover, Replay and Relay threats were identified as significant, pointing to the manipulation of signal or data transfer as critical areas of risk. This coherent analysis underscores the strategic focuses of attackers and underscores the need for robust defenses in these identified areas.

Based on the analysis, several insights and critical areas of focus in the data were categorized into six overarching themes, as can be seen in Table 6.

	Themes	Codes	Assets Affected	Consequences
1	Signal Interference and Manipulation	Jamming, Spoofing, Meaconing, Replay, Relay and Key Fob/Keyless entry Jamming	GPS/GNSS, LiDAR, RADAR, Key fob/Keyless entry	Loss of signal lock, Misleading position or timing data, Inaccuracies in location determination and Vehicle theft.
2	Physical Manipulation and Deception	Injection of Fake Obstacles, Hidden Objects, Electromagnetic Interference (EMI)	LiDAR, RADAR, Camera	Creation of non-existent obstacles, Inability to detect real obstacles, Incorrect object detection or categorization.
3	Unauthorized Access and Control	Cryptographic Analysis Attacks; Elevation-of-Privilege	ECUs, Voice Controllable Systems (VCSs)	Vehicle theft, Unauthorized engine start, Unauthorized control over vehicle systems.
4	Data Integrity and Availability	Tampering, Denial of Service, Repudiation, Software Flashing	Camera, ECUs, TPMS, Onboard Diagnostic Port (OBD)	Inaccurate crash data, Compromised emergency responses, Disrupted communications, Misleading information.
5	Communication and Network Vulnerabilities	Spoofing CAN Protocol, LIN Bus Error-Handling Mechanism, Spoofing	CAN, LIN, FlexRay, Ethernet	Manipulation of vehicle functions, Unauthorized access to vehicle network, Inject

	Themes	Codes	Assets Affected	Consequences
		FlexRay Protocol, Close Proximity Vulnerabilities, ARP cache poisoning, Denial of Service, SOME/IP Spoofing and TCP Hijacking	and Bluetooth devices.	unauthorized frames into the network, Exploitation or disruption of the system.
6	External Network Exploits	Spoofing, Tampering and Malware Injection by Privilege Accessing in Over-the-air (OTA) Updates, Command Injection Attack, Network Eavesdropping.	Backend operational, Systems. Cloud Computing. Wi-Fi and Cellular network	Loss of control over ECUs, Blockage of critical updates, Potential takeover of the vehicle, Remote access to the vehicle's systems from anywhere without physical interaction.

Table 6 Overview of themes and codes derived from document analysis

1) Signal Interference and Manipulation

This theme emphasizes the vulnerability of AV communication systems to disruption and deceit. Threats like Jamming, Spoofing, Meaconing, and Replay directly impact the AV's ability to accurately receive and interpret signals from GPS/GNSS, LiDAR, and RADAR. Jamming refers to the deliberate transmission of radio frequency signals to disrupt the communication channels, overpowering the genuine signals with noise or other forms of interference. In the context of AVs, jamming can cause a vehicle to lose connection with satellite navigation systems like GPS/GNSS, leading to a loss of location services. A more insidious form of interference, Spoofing involves the creation of fake signals that mimic legitimate ones. For AVs, this might mean creating false GPS/GNSS signals that mislead the vehicle's navigation system into believing it is in a different location. Similar to spoofing, meaconing involves capturing legitimate signals (for instance, from GPS/GNSS satellites) and retransmitting them with a delay. This delayed signal can cause miscalculations in the AV's positioning systems, leading to errors in navigation and timing, potentially resulting in unsafe maneuvers or decisions by the vehicle. Replay involves capturing a legitimate signal and replaying it at a later time or different location. For AVs, a replay attack could involve retransmitting previously recorded GPS/GNSS signals to the vehicle, causing it to receive outdated or location-irrelevant positioning information. The consequences, such as loss of signal lock and misleading position data, can severely compromise navigation and situational awareness, potentially leading to unsafe operational conditions.

This theme also highlights the use of signals to gain unauthorized access to the vehicle. Attackers use signal jamming to prevent the key fob's locking signal from reaching the vehicle, leaving it unlocked and susceptible to theft or unauthorized entry. This method is simple but effective and exploits the reliance on wireless signals for locking and unlocking, posing a significant security risk. Attackers also use Replay Attacks to capture the signal sent by a key fob to the vehicle and replay it to unlock the vehicle later. While modern vehicles often use rolling codes that change with each use to counter this threat, attackers can still find ways to circumvent these protections, particularly if they can block the original signal and force the key fob to transmit multiple codes. Besides, Relay Attacks are used to bridge the distance between the key fob and the vehicle, attackers can trick the vehicle into thinking the key fob is closer than it actually is. This type of attack is particularly concerning for keyless entry and start systems, as it can allow for vehicle theft without needing to physically access the key fob.

2) Physical Manipulation and Deception

Physical manipulation and the introduction of deceptive inputs pose a significant risk to the perceptual accuracy of AVs. Threats can trick AV sensors into seeing false obstacles, such as projected images or shapes that LiDAR or cameras mistake for real objects. This threat can cause AVs to slow down, stop, or swerve unnecessarily, creating potential road hazards. Conversely, techniques that make real objects "invisible" to sensors also pose risks. Materials absorbing LiDAR signals or designs deflecting RADAR waves can hide actual obstacles or vehicles, increasing the risk of collisions if the AV does not react properly.

However, it is not physical manipulation in the traditional sense, there is EMI can physically alter the operational capacity of AV sensors. Deliberate electromagnetic interference aimed at LiDAR's time-of-flight components can result in inaccurate readings or failure to detect objects correctly. This theme highlights the importance of developing advanced detection algorithms capable of distinguishing between genuine and fabricated inputs, enhancing the AV's ability to accurately interpret its surroundings and make safe driving decisions.

3) Unauthorized Access and Control

This theme reveals the AV's susceptibility to breaches that could lead to vehicle theft and unauthorized manipulation of vehicle systems. It highlights the importance of securing key fobs, ECUs, TCUs, and voice-controllable systems against sophisticated attacks designed to gain unauthorized access or control. Sophisticated techniques like cryptographic analysis and side-channel attacks are particularly concerning. These methods target the cryptographic algorithms used in vehicle communication systems and exploit vulnerabilities in encryption. By analyzing variations in power consumption, timing, and electromagnetic emissions from ECUs or key fobs, attackers can potentially decipher secret keys, gaining unauthorized access or control. Additionally, the finding also reveals that voice controllable systems are at risk from hidden voice command attacks like Dolphin Attack and LipRead, which can lead to unauthorized actions or access. This vulnerability underscores the need for robust protection measures for all critical components, including key fobs, which directly interface the vehicle's security system.

4) Data Integrity and Availability

AVs face several significant security threats, including tampering, Denial of Service (DoS), repudiation, and risks from software flashing and malware. Tampering, which entails unauthorized alterations to system data or functionality, can deceive the AV's decision-making, directly endangering safety. DoS attacks, aimed at overwhelming system resources, can disable critical functions and disrupt internal or external communications, compromising safety. Repudiation involves disputing the legitimacy of actions, challenging the authenticity of crucial data like command signals or sensor readings, and thus complicating accountability. Software threats, including flashing and malware, pose risks by introducing harmful functions or disabling essential systems, affecting both vehicle performance and data security.

Ensuring the integrity and availability of data and services is paramount in AV systems, as compromised data can lead to inaccurate analytics, delayed emergency responses, and disrupted communications. This theme points to the necessity of robust data verification and validation processes, secure data transmission protocols, and reliable backup systems to maintain the accuracy and reliability of the data used by AVs for decision-making and reporting.

5) Communication and Network Vulnerabilities

This theme focuses on the vulnerabilities within vehicle communication networks, including CAN, LIN, FlexRay, Ethernet and Bluetooth devices. The CAN bus, essential for integrating microcontrollers and devices without a central computer, lacks critical security features like authentication and encryption, making it prone to attacks that could disrupt essential functions such as braking and steering. Similarly, the LIN bus, handling simpler tasks like window and mirror controls, shares this vulnerability due to its absence of advanced security measures, leaving it open to malicious commands. Despite FlexRay's higher data rate and use in critical applications, it too faces risks of spoofing and tampering, underscoring that advancements in data handling do not inherently resolve security issues. Ethernet, increasingly crucial in AVs for its high-speed data transfer and real-time communication capabilities, faces significant threats such as ARP cache poisoning, DoS attacks, SOME/IP spoofing, and TCP hijacking. These vulnerabilities can lead to unauthorized access, data manipulation, and disruption of network communication, posing substantial risks to the security and functionality of AVs.

Furthermore, Bluetooth and short-range wireless technologies, crucial for keyless entry, infotainment, and connectivity, are susceptible to eavesdropping, spoofing, and relay attacks, posing threats to both vehicle security and personal data integrity.

6) External Network Exploits

The susceptibility of AV systems to external network exploits, particularly through OTA software or firmware updates, presents a considerable risk. However, if not securely managed, they can become a vector for attackers to introduce malicious code or unauthorized modifications. Threats include man-in-the-middle attacks where attackers intercept and alter the update before it reaches the vehicle or attacks on the update server itself, compromising the integrity of the updates. Social Engineering attacks are used to manipulate individuals within the organization or supply chain into disclosing confidential information or performing actions that compromise security. For AVs, this could mean tricking employees into providing access to secure networks, revealing cryptographic keys, or bypassing security protocols, thereby granting attackers access to vehicle systems or sensitive data.

Furthermore, remote access attacks to AV systems frequently involve complex interactions with external networks, encompassing not only connections to OEM back-end servers but also vehicle-to-infrastructure (V2I) communications. These pathways are indispensable for the functioning and updating of AV systems, but they are also susceptible to exploitation by attackers. Vulnerabilities in communication protocols and software interfaces for remote interactions are primary targets.

Researchers at Tencent Keen Security Lab demonstrated significant cybersecurity threats to AVs by exploiting vulnerabilities in the Tesla Autopilot Electronic Control Unit (APE). By remotely accessing the APE through external network connections such as 3G/Wi-Fi, they managed to gain root access and manipulate the steering system from a remote mobile device. This critical breach allowed them to control the vehicle's steering, highlighting the profound implications of such vulnerabilities. Additionally, they used adversarial examples to disrupt the autowipers function and subtly modified road markings to mislead the Tesla into entering the wrong lane, illustrating the diverse ways external network vulnerabilities can be exploited.

In another notable instance, one of the most infamous hacks involving automotive systems was the remote compromise of a Jeep Cherokee. Researchers capitalized on the vehicle's cellular connection via the Uconnect system, which communicates over Sprint's cellular network. They targeted this connection to gain remote access to the vehicle's systems from any location without needing physical interaction. This case underscores the broad-reaching implications of exploiting such network vulnerabilities, as attackers can remotely access and control critical vehicle functions such as steering and braking at speed, posing significant safety risks.

5.1.2 Threats Identified through TARA Methodology

The TARA methodology commences by identifying the assets within the automotive system. For this study, due to time constraints necessitating a focused analysis, four key components of the AV system were chosen for analysis within the scope of this study: **the CSM, sensors, VCM, and the Backoffice System.**

When identified threats using TARA methodology within this research, specific assumptions regarding sensors are delineated in Table 7. The purpose of the table is to highlight that a single compromised sensor may not significantly affect the vehicle's perception system, while multiple compromised sensors could lead to various damage scenarios. This reason is because AVs rely on sensor fusion techniques to create a comprehensive understanding of their surroundings. If more than one sensor is compromised, the vehicle's ability to accurately perceive its environment is more severely degraded, potentially leading to incorrect actions that can result in various levels of damage or risk. Following is an explanation for each threat outlined in the table.

Threat 1: With all perception systems compromised, the vehicle would essentially be navigating without any sensory input to validate its environment, making safe driving impossible. This threat scenario could lead to loss of control of critical functions.

Threat 2: This scenario considers the simultaneous compromise of both GNSS and IMU systems, which are integral to the AV's navigation and orientation capabilities. Since these systems tend to corroborate each other's data, an attacker must disrupt both to effectively mislead the vehicle's sense of position, potentially leading to loss of control of critical functions.

Threat 3,4,5: An attacker would need to manipulate two out of three sensors, Radar, LiDAR, and camera, to significantly impact the vehicle's vision, potentially leading to loss of control of critical functions.

Sensors	Threat 1	Threat 2	Threat 3	Threat 4	Threat 5	Threat 6
GNSS	x	x				
IMU	x	x				
RADAR	x		x	x		x
LiDAR	x		x		x	x
Camera	x		x	x	x	
Damage Scenario	*DS_5 Loss of control of critical functions*	*DS_9 Loss of remote control of the vehicle*	*DS_5 Loss of control of critical functions*	*DS_5 Loss of control of critical functions*	*DS_5 Loss of control of critical functions*	*DS_5 Loss of control of critical functions*
Assumption		Assume that GNSS and IMU provide similar data and an attacker need to compromise both of them to munipulate the position of the vehicle.		Assume that an attacker need to compromise 2 out of 3 (RADAR, LiDAR,Camera) to munipulate the vision of the vehicle.	Assume that an attacker need to compromise 2 out of 3 (RADAR, LiDAR,Camera) to munipulate the vision of the vehicle.	Assume that an attacker need to compromise 2 out of 3 (RADAR, LiDAR,Camera) to munipulate the vision of the vehicle.

Table 7 The specified conditions in the threat analysis of sensors

In Table 8, cybersecurity threats are delineated using the TARA methodology, aligning with the themes outlined in section 5.1.1. Each theme is attributed with distinct codes, reflective of the nuances and attributes of the threats embedded within that theme. These codes originate from threat scenarios crafted during the attack analysis phase. The consequences associated with each threat scenario are derived from the damage scenarios outlined during the TARA process.

	Themes	**Codes**	**Assets Affected**	**Consequences**
1	Signal Interference and Manipulation	Spoofing the satellite to GNSS	Sensors (GNSS)	Loss of remote control of the vehicle
		Spoofing environment to sensors	Sensors (LiDAR, RADAR and camera)	Loss of remote control of the vehicle, Loss control of the critical functions (active safety)
		Launching DoS to sensors signals using Transmitters or Jammers	Sensors (GNSS, IMU, LiDAR, RADAR and camera)	
2	Physical Manipulation and Deception	Tampering with sensors, Physical Interference with sensors	Sensors (GNSS, IMU, LiDAR, RADAR and camera)	Loss of remote control of the vehicle, Loss control of the critical functions (active safety)
3	Unauthorized Access and Control	Spoofing key fob signal, Cloning key fob Software, Signal relay attack	CSM	Vehicle theft, Vehicle is not able to start the engine (immobilizing the vehicle)
		Unauthorized access to CSM software and read symmetric key used for authentication of the key fob		Vehicle theft
4	Data Integrity and Availability	Tampering CSM software via OBD port access, Tampering CSM software via the Backoffice System access, Launching DoS attack on CSM software, Launching DoS Attack on CSM software via physical access	CSM	Vehicle is not able to start the engine (immobilizing the vehicle), Vehicle theft

	Themes	Codes	Assets Affected	Consequences
		Tampering sensors software via OBD port access, Tampering sensors software via the Backoffice System access, Launching DoS attack on sensors software, Launching DoS Attack on sensors software via physical access	Sensors	Loss of remote control of the vehicle, Loss control of the critical functions (active safety)
		Tampering with VCM Software via OBD Port Access, Tampering with VCM Software via Backoffice System Access, Launching DoS Attack on VCM Software via Physical Access, Launching DoS Attack on VCM Software via Wireless Interception	VCM	Vehicle is not able to start the engine (immobilizing the vehicle), Passenger's data breach, Loss control of the critical functions (active safety), Loss control of the non-critical functions, Loss of passive safety functionalities, Remote control of the vehicle by the attacker
5	Communication and Network Vulnerabilities	Spoofing ECUs to VCM by manipulating CAN bus messages	VCM, CSM	Loss control of the non-critical functions, Loss control of the critical functions (active safety), Loss of passive safety functionalities, Vehicle is not able to start the engine (immobilizing the vehicle), Vehicle theft
6	External Network Exploits	Tampering VCM by injecting malicious code during remote software update	VCM	Loss of connectivity, Vehicle theft, Vehicle is not able to start the engine (immobilizing the vehicle), Passenger's data breach, Loss control of the critical functions (active safety), Loss control of the non-critical functions, Loss of passive safety functionalities, Remote control of the vehicle by the attacker
	Remark: Upon analysis of the Backoffice System, no threat scenarios have been identified			

Table 8 Overview of themes and codes derived from conducting TARA

1) Signal Interference and Manipulation

This theme highlights the vulnerability of AV sensor systems to manipulation and interference. The research reveals sophisticated attack vectors aimed at disrupting the integrity and reliability of sensor data, posing significant challenges to the safe operation of AVs.

One prominent threat within this theme is GNSS spoofing attacks, which involve the generation of counterfeit signals designed to deceive the vehicle's navigation systems. Attackers can either generate non-coherent signals to force the vehicle's GNSS receiver into search mode or produce signals closely resembling genuine GNSS signals to override authentic ones, thereby compromising the accuracy of the vehicle's navigation.

The use of LiDAR, RADAR, and camera sensors introduces new opportunities for manipulation. Attackers can intercept radio waves emitted by RADAR sensors or laser signals emitted by LiDAR sensors to generate false signals, thereby manipulating the sensor's perception of the environment. Additionally, attackers can modify signals to inject false objects into the sensor's perception or remove existing objects from detection, compromising the vehicle's ability to accurately perceive its surroundings. Another tactic employed by attackers involves the creation of visually deceptive objects using various technologies such as monitors, projectors, holographic projection devices, or electronic

billboards. By manipulating camera sensors, attackers can deceive the vehicle's perception systems, leading to potentially hazardous situations.

Attackers may also utilize transmitters or jammers to interfere with sensor signals, introducing random or structured noise or overpowering genuine signals. Physical manipulation of sensors by altering the surrounding environment can also be used to cause a DoS scenario.

The consequences of these attacks can be severe, potentially resulting in the loss of remote control of the vehicle or loss of control over critical functions, particularly in scenarios where multiple sensors are compromised simultaneously, as assumed in Table 10.

2) Physical Manipulation and Deception

This theme reveals a complex threat landscape surrounding the physical manipulation and deception of sensor components in autonomous vehicles. Attackers exploit vulnerabilities by accessing the vehicle and tampering with sensors such as GNSS, IMUs, LiDAR, RADAR, and cameras. This tampering involves replacing genuine sensors, manipulating calibration parameters, or interfering directly with hardware, leading to false sensor readings. Consequently, the vehicle's perception systems are compromised, resulting in loss of remote control and critical function control, especially in active safety features. An attacker also employs physical interference tactics, such as shaking sensors or using external light sources, to disrupt sensor operation and distort readings further. This threat undermines the vehicle's ability to accurately perceive its surroundings and maintain remote control, posing significant risks of losing remote control of the vehicle and losing control of critical functions.

3) Unauthorized Access and Control

This theme delves into the issue of unauthorized access and control within AVs. Within the scope of assets analyzed in this study, vulnerabilities are identified in the CSM. Attack methods employed under this theme are exploit weaknesses in key security mechanisms. These methods include spoofing key fob signals, cloning key fob software, relaying signals from legitimate key fobs to the vehicle, and gaining unauthorized access to CSM software to obtain symmetric keys used for key fob authentication. The consequences of these unauthorized access and control threats are the risk of vehicle theft, where attackers exploit vulnerabilities in keyless entry systems to gain unauthorized access to the vehicle and potentially steal it. Additionally, these threats can lead to the immobilization of the vehicle, rendering it inoperable and causing disruption to the owner or user.

4) Data Integrity and Availability

This theme addresses the challenge of ensuring that data within the vehicle's systems remains accurate, reliable, and accessible. Maintaining data integrity ensures that information stored and transmitted by various components of the vehicle is trustworthy and unaltered. Similarly, data availability ensures that the required information is accessible when needed, without any disruptions or delays. These vulnerabilities could stem from software flaws, insecure communication protocols, or inadequate access controls.

Attackers employ various methods to compromise data integrity and availability. For instance, they may tamper with software running on these components through the OBD port, which provides access to vehicle diagnostics and configuration settings. They may exploit vulnerabilities in back-office systems connected to the vehicle network to manipulate software remotely. Furthermore, they may launch DoS attacks targeting these components, aiming to overwhelm them with excessive traffic or requests, thereby disrupting their normal operation.

The consequences of such attacks can be severe and wide-ranging. For the CSM, compromised integrity and availability of data can lead to vehicle theft or immobilization by preventing the engine from starting. Tampering with sensors can result in loss of remote control over the vehicle and compromise critical safety functions, such as collision avoidance systems. In the case of the VCM, the impact can

be even more significant, including loss of connectivity, potential vehicle theft, engine immobilization, breach of passenger data, loss of control over critical and non-critical functions, and susceptibility to remote control by malicious actors.

5) Communication and Network Vulnerabilities

This theme focuses on exploitable weaknesses within the vehicle's communication networks. For instance, the attacker gains physical access to the vehicle's internal network by accessing the wiring harnesses and using specialized equipment such as a CAN injector. This device is capable of manipulating communication on the vehicle's CAN bus network, which is used for inter-device communication within the vehicle. Once connected to the internal network, the attacker can send fake or altered messages directly to ECUs within the vehicle that communicate with the VCM. These messages are intended to mislead the VCM, the central control unit of the vehicle, into making incorrect decisions or taking inappropriate actions. The consequences of these attacks may result in loss of control over critical and non-critical functions, loss of passive safety functionalities, immobilization of the vehicle, and potential vehicle theft.

6) External Network Exploits

This theme highlights the vulnerability of AV to external network exploits, particularly during remote software updates. An attacker intercepts communication between the cloud-based update server and the VCM, which is a critical component responsible for controlling various vehicle functions. By intercepting this communication, attackers can inject malicious code into the software update process. This injected code could potentially compromise the integrity of the vehicle's control systems, leading to severe consequences such as loss of connectivity, vehicle theft, engine immobilization, passenger data breaches, loss of control over critical and non-critical functions, loss of passive safety functionalities, and remote control by attackers, underscore the severity of the vulnerabilities associated with external network exploits. These consequences highlight the wide range of risks posed by such exploits, including both immediate threats to vehicle functionality and broader concerns about passenger safety and data security.

5.1.3 Comparing the results of document and TARA Methodology

Based on the threats identified in sections 5.1.1 and 5.1.2, the research found that the TARA Methodology and document analysis reveal both similar and distinct threats. Table 9 illustrates these findings, offering a structured comparison of the types of threats and the specific assets or systems they impact within each identification method.

	Threat Category	Identified threats in 5.1.1 (Document Analysis)	Identified threats in 5.1.2 (TARA Methodology)
1	Signal Interference and Manipulation	Jamming, Spoofing, Meaconing which affects GPS/GNSS, LiDAR, RADAR and Key fob or keyless entry	Spoofing satellite to GNSS, environmental spoofing, DoS using jammers on sensors
2	Physical Manipulation and Deception	Injection of fake obstacles, EMI affecting LiDAR, RADAR, Cameras	Tampering and physical interference with sensors like GNSS, IMU, LiDAR
3	Unauthorized Access and Control	Cryptographic attacks, Elevation-of-privilege on ECUs and VCSs	Spoofing/cloning key fob, unauthorized access to CSM software
4	Data Integrity and Availability	Tampering, DoS, repudiation, software flashing on camera, ECUs	Tampering and DoS on CSM, sensor, and VCM software via various access points
5	Communication and Network Vulnerabilities	Spoofing, vulnerabilities in CAN, LIN, FlexRay, Ethernet, Bluetooth	Spoofing ECUs to VCM by manipulating CAN bus messages
6	External Network Exploits	Tampering, Spoofing, Malware injection by Privilege Accessing during OTA update process or via backend systems, cloud, Wi-Fi, cellular networks	Tampering VCM by injecting code during remote updates

Table 9 Comparing result of document and TARA Methodology

To discern the unique identified threats between sections 5.1.1 and 5.1.2, the researchers examined threats exclusive to each section based on their context and details and this research found that:

1) Unique Threats Identified in 5.1.1 (Document Analysis), but later excluded

- Key Fob/Keyless Entry Jamming: This threat targets the vehicle's security system through signal interference.
- Spoofing, Tampering, and Malware Injection in external network: These threats are performed via cloud computing, backend operational systems, and broader Wi-Fi and cellular networks. They can lead to loss of control over Engine Control Units (ECUs) and other critical vehicle functions.

This research also identified both threats during the risk assessment phase of the TARA. However, these threats were excluded from the final TARA documentation. This decision was based on specific architectural assumptions made in the analysis, which made these types of attacks highly improbable or practically impossible.

2) Unique Threats Identified in 5.1.2 (TARA Methodology)

- Tampering with VCM Software: This involves accessing the software remotely through back-office systems to alter vehicle control.
- Spoofing ECUs to VCM: This threat emerge through physical access to the vehicle's internal network, specifically manipulating CAN messages.
- Cloning Key Fob Software: This involves duplicating the software of a vehicle's key fob to gain unauthorized access.
- Unauthorized Access and Reading Symmetric Keys: Attackers physically access the key fob to extract symmetric keys used for the vehicle's authentication process.

Thus, this comparative analysis highlights the critical importance of understanding the specific contexts in which these threats occur to enhance AV security measures effectively. Besides, the comparison between sections 5.1.1 and 5.1.2 in terms of identified threats reflects the different methodologies and potential limitations inherent to each approach.

1) **Scope and Depth:** This research conducted document analysis covers a wider scope but with less depth in specific vulnerabilities, while conducting TARA provides a narrower but deeper insight into specific threats and their operational impacts.
2) **Operational and General Security:** Document analysis might miss some specific operational vulnerabilities due to its broader focus, whereas TARA's targeted approach might not capture the full breadth of the threat landscape as seen in document analysis. Because TARA is asset-centric, it emphasizes threats that directly impact the chosen assets. Thus, TARA may miss more general or emerging threats that have not yet been identified as specifically targeting those assets. Broader threats, such as new malware variants, emerging cyberattack trends, or vulnerabilities introduced by interconnected systems, might not be captured if they have not been mapped to the selected assets in the preliminary stages of the TARA process.
3) **Direct or indirect threat:** In AVs, threats can arise not just from direct attacks on critical systems but also from less direct sources, such as vulnerabilities in third-party components, backend systems, or even user interfaces. Document analysis, by surveying a wider range of sources and types of information, often captures these broader, sometimes more speculative threats that a targeted TARA might not consider if they do not immediately impact the primary assets being assessed.
4) **Rapidly Changing Threat Landscape:** The cyber threat landscape evolves quickly, with new vulnerabilities and attack methods developing constantly. TARA's in-depth focus is excellent for current and well-understood threats but may not adapt quickly enough to include the latest threats identified in broader research or industry-wide alerts unless the methodology is regularly updated to include these new threats.

While the TARA methodology successfully captured all threats identified in the document analysis, it is essential to recognize that TARA's focused, asset-centric approach may not encompass the entire spectrum of emerging or indirect threats identified in broader document analysis. This highlights the effectiveness of TARA in identifying and addressing relevant threats within the AV context but also underlines the need for continuous updates and broader research integration to capture the rapidly evolving threat landscape.

5.1.4 Evolving threat landscape of AVs

The evolving threat landscape of AVs is increasingly complex and multifaceted, reflecting the sophisticated nature of systems and their integration with cutting-edge technologies for navigation, decision-making, and operation in real-time. Based on the findings from this research, the landscape encompasses a range of cyber threats that exploit the unique vulnerabilities of AVs, particularly those related to their reliance on connectivity, external signals, and advanced sensors for navigation and operation.

AVs are heavily reliant on external signals such as GPS/GNSS, and sensor data from LiDAR and RADAR for navigation and interpreting their environment. This reliance introduces vulnerabilities where malicious actors can disrupt these signals through jamming, spoofing, and other forms of interference. Thus, attacks can lead to loss of signal lock, misleading data being fed into the AV's navigation system, and, ultimately, unsafe operational conditions. Furthermore, the threat actors have moved beyond digital interference to physically manipulate the environment around AVs. By creating fake obstacles or making real objects invisible to AV sensors, attackers can deceive the vehicle's perception systems. This could cause AVs to make erroneous navigational decisions, such as stopping for non-existent obstacles or failing to recognize actual hazards, posing significant safety risks. While the digital nature of AV operations, coupled with their complex communication systems, still opens up avenues for unauthorized access and control. The threat vectors like key fob signal spoofing, cloning, and sophisticated cryptographic analysis attacks aim at gaining unauthorized access or control over the vehicle systems. In this case, threats not only raise concerns about vehicle theft but also unauthorized control over AV operations.

Moreover, AVs rely on a continuous stream of accurate and timely data for decision-making processes critical to safe navigation and operational efficiency. However, this reliance also presents a vulnerability. Threats such as tampering, DoS attacks, and unauthorized software flashing directly challenge the sanctity of this data. When data integrity and availability is compromised, the consequences can be immediate and severe, ranging from erroneous decision-making to total system failure, underscoring the paramount importance of securing data against such threats. In addition, the seamless operation of AVs is contingent upon complex communication networks that facilitate the exchange of data between the vehicle's internal systems and the external environment. This network includes protocols and interfaces like CAN, LIN, FlexRay, Ethernet and Bluetooth devices. Unfortunately, these networks are not impervious to breaches. They can be exploited for malicious purposes, including vehicle function manipulation, unauthorized network access, and malicious code execution. This attack surface includes vulnerable over-the-air (OTA) software update mechanisms, which are compromised through man-in-the-middle attacks or social engineering and can serve as conduits for widespread system compromises. Such attacks can alter the very firmware of AVs, introducing malicious functionalities or disabling critical systems.

In investigating the threat landscape of AVs, this research identifies several threats that are particularly pronounced due to the unique architectural and operational features of these advanced vehicles. Unlike traditional vehicles, AVs incorporate a range of technologies and systems, while enhancing their capabilities and also introducing specific vulnerabilities. The following outlines unique threats in AVs from the findings.

1) AVs' heavy dependence on external signals and connectivity, such as GPS/GNSS, LiDAR, and RADAR, for navigation and operational decisions, presents a significant risk of signal interference

and manipulation. This reliance is a direct result of the autonomous nature of these vehicles, which contrasts with the more limited use of such technologies for basic functions in non-AVs.

2) The deployment of advanced sensor arrays in AVs for environmental perception and situational awareness introduces vulnerabilities unique to their mode of operation. These vulnerabilities encompass risks of physical manipulation and deception that can compromise the vehicles' ability to accurately interpret their surroundings, challenges that are inherently linked to the sophisticated sensor technologies utilized by AVs.
3) AVs' advanced automated control systems elevate the risks associated with unauthorized access and control. Self-driving vehicles are equipped with advanced systems that make driving decisions without human input. These systems rely on digital keys and secure networks for their operations. The primary concern is the risk of unauthorized access. If hackers manage to bypass these security measures, they could take control of the vehicle. This risk is unique to AVs since traditional vehicles require manual control by a human.

The findings also identify damages from the cybersecurity threats in AVs and reveal a diverse and evolving landscape of vulnerabilities that lead to significant damages, impacting both the vehicles' operational integrity and the safety of passengers. Signal interference undermines navigation through physical attacks that deceive perception systems into making hazardous decisions. Unauthorized access via methods like key fob hacking poses risks of vehicle theft and compromises passenger safety by allowing external control over the vehicle's critical functions. Furthermore, data integrity breaches threaten decision-making processes and emergency responses, while network vulnerabilities open pathways for attackers to manipulate vehicle operations or even assume complete control, especially through OTA updates.

However, the exploration of cybersecurity threats to AVs in this research has identified key areas where the integration of advanced technologies and reliance on automated systems introduce specific damages not applicable to traditional, non-AVs. The research emphasizes the unique challenges faced by AVs in terms of operational safety, data security, and vehicle control, underlining how these challenges are magnified by the autonomous nature of these vehicles.

Operational safety is particularly critical in AVs due to their dependence on sensors and signals for navigation and obstacle detection. In contrast to non-autonomous vehicles, which primarily rely on human perception and decision-making, AVs' reliance on digital inputs makes them susceptible to interference and manipulation threats, posing unique safety risks.

Furthermore, data security becomes paramount in the context of AVs, as they operate on complex data algorithms for decision-making processes. This research highlights that any compromise to data integrity not only poses a cybersecurity risk but directly impacts the safety and reliability of AV operations, a concern less pronounced in non-AVs where decision-making is less dependent on digital data. Moreover, vehicle control represents an area where AVs encounter distinct challenges. The potential for unauthorized access and control presents a significant concern, given AVs' ability to operate autonomously, without human intervention. This scenario introduces an elevated risk of theft or malicious control.

The evolving threat landscape in AVs presents a complex and multifaceted challenge. As AV technology advances rapidly, so too do the tactics and motivations of threat actors seeking to exploit vulnerabilities in these systems. Given that AVs represent some of the most cutting-edge technology in the automotive industry, they naturally attract the attention of threat actors.

The primary motivation for many hackers remains financial gain. Individual hackers, leveraging their skills and choose to operate as black hats, exploiting vulnerabilities in AV software or infrastructure to manipulate or disable essential functions. Gaining unauthorized access to vehicle functions to temporary or permanent loss of control over AVs. In addition to manipulating vehicle functions, black hats may also exploit data vulnerabilities in AV systems to steal sensitive information. Personal information from passengers, such as location data or contact details, can be targeted by hackers for illicit purposes, including identity theft, fraud, or surveillance. One tactic to hack for financial gain is, the trend of hacking to unlock paid features for free, which typically require activation through payment or subscription, is gaining traction in the AV industry [94]. However, not all hackers have malicious

intent. AV technology has also brought about a rise in white hat hackers, driven by a desire to improve cybersecurity and protect AV systems from exploitation while also seeking financial rewards. Companies like Tesla actively encourage this approach by offering substantial rewards for identifying vulnerabilities in their systems. For instance, Tesla hackers have been rewarded with prizes $200,000 cash and even a brand-new Model 3 for uncovering new vulnerabilities [95]. This incentivizes a shift from portraying hackers as the "bad guys" to recognizing them as valuable contributors to enhancing AV security.

5.2. Evaluate TARA methodology when applied to address the cybersecurity threats in AVs

In this section, the research aims to assess the effectiveness of the TARA methodology within the framework of ISO/SAE 21434 in addressing cybersecurity threats in AVs. The research question guiding this evaluation is: "*What are the limitations and strengths of the TARA methodology under the framework of ISO/SAE 21434 when applied to address cybersecurity threats in autonomous vehicles?"*

1) STRIDE Threat Modeling

In the initial phase of this research's risk assessment, the threat analysis process was conducted using the STRIDE threat modeling approach of four specific components. The CSM, Sensors, VCM, are categorized as processes within the system. The Backoffice System is categorized as an external entity in relation to the E/E system.

The methodology presents a strength in its ability to guide a structured identification of threats, linking them to specific cybersecurity assets and scenarios. The detailed mapping of STRIDE model and threat scenario showcases TARA's capacity to dissect and categorize complex systems into manageable units of analysis. The methodology captures a wide range of threats from "Spoofing" to "Denial of Service", demonstrating the comprehensiveness of TARA in considering various attack vectors.

While conducting threat analysis within the scope of AVs, limitations become apparent when applying the STRIDE to analyze "environment". Theoretically, STRIDE is a framework designed to analyze threats within interactions between two elements of a system, such as between software components or between a user and a system. Given that the environment acts as an external entity (see Figure 5) interacting with sensors, it introduces unique challenges that may not neatly align with the traditional STRIDE model.

As shown in table 10. In the threat scenario, the term "spoofs sensors to environment" means to impersonating sensors to deceive the environment. However, sensors typically do not actively interact with or transmit data to the environment in the same manner that systems interact with each other. In the context of AVs, sensors typically operate on a one-way communication model. They collect data from the environment but do not transmit data back to the environment. Sensors such as LiDAR, radar, and cameras capture information about the AV's surroundings but do not actively send data to external entities like the environment. Therefore, the scenario of spoofing sensor signals to the environment may not be applicable.

Similarly, in the scenario of "repudiation" with the environment, an attacker may attempt to repudiate their interactions or actions within the environment, leading to challenges in attribution or accountability. Repudiation threats typically involve denying or disowning actions or interactions within a system. In traditional cyber this might involve denying actions taken by a user or tampering with logs to erase evidence of certain activities. However, when an attacker manipulates the environment, attributing these actions directly becomes significantly more challenging. This is because the environment serves as the backdrop against which these actions occur, making it challenging to attribute them accurately.

Asset	Threat Category	Threat scenario	Remark
environment	spoofing	An attacker spoofs sensors to environment	Not applicable
environment	repudiation	An attacker repudiates environment.	Not applicable

Table 10 Threat analysis of the environment

However, as shown in table 11, when analyzing threats to sensors component in AVs, it is imperative to consider the environment. For instance, threat scenario of an attacker spoofing environment to sensors can introduce significant vulnerabilities. An attacker may attempt to deceive sensors by spoofing environmental data. They could manipulate RADAR signals to create false detections or cause the sensor to miss targets by emitting deceptive signals. Similarly, the attacker could manipulate LiDAR signals by emitting false laser pulses or altering reflected signals, leading the sensor to misinterpret the environment. These actions can compromise the accuracy of sensor readings, potentially resulting in erroneous perceptions of the surrounding environment by the system.

Asset	Threat Category	Threat scenario	Remark
sensors	Spoofing	An attacker spoofs environment to sensors.	

Table 11 Threat analysis of sensors

The STRIDE model, while comprehensive in its coverage of various cybersecurity threats, may not seamlessly accommodate threat analysis of environment within the discrete categories it defines. Analyzing the environment itself might not neatly fit into the STRIDE model's framework. However, when considering the environment in conjunction with sensors, it becomes imperative to assess the dynamic interplay between environmental factors and sensor functionality. These dynamic and non-deterministic aspects of environmental threats may pose challenges when attempting to categorize them within the structured framework of the STRIDE model as it may not fully capture the nuances and complexities of environmental threats encountered in AVs systems.

2) Expertise Challenge in Feasibility Analysis

TARA Feasibility ratings in AV systems involve assessing the chance of potential cyber-attacks and their potential impact. However, this process is partially subjective and uncertain due to the evolving nature of technology and the diverse range of stakeholders involved.

The threats targeting AV sensors represent a significant departure from traditional cyber threats, as they exploit vulnerabilities inherent in the physical operation of these sensors rather than targeting software or network infrastructure. Consider, for example, a scenario where an attacker launching DoS to cameras, LiDAR, or RADAR sensors using unconventional methods like high-intensity LED lights or powerful flashlights does not rely on exploiting code or network weaknesses but rather on manipulating physical phenomena such as light propagation and sensor response.

Gaining insights into sensor vulnerabilities and the feasibility of attacks may requires integrating physics knowledge into cybersecurity practices. This includes understanding concepts such as light interference, signal attenuation, and sensor noise, which directly impact the reliability and security of AV sensor systems.

5.3. Mitigation Strategies to Threats in AVs

This section explores mitigation strategies for cybersecurity threats to AVs. It begins by analyzing strategies from documented research to address AV vulnerabilities. The section then shifts focus to the

evolving threat landscape, highlighting key vulnerabilities like signal interference, physical tampering, and unauthorized access, ensuring a comprehensive approach to AV cybersecurity.

5.3.1 Mitigation strategies identified through documents

In the evolving and complex landscape of AV technology, cybersecurity is increasingly vital. As the integration of advanced electronics and software enhances AV capabilities, it also opens up numerous vulnerabilities to cyber threats. To effectively counter these vulnerabilities, this research adopts a structured approach through Thematic Mitigation table to answer the research question: *What are mitigation strategies to address threats in autonomous vehicles?* This table organizes mitigation strategies into key themes based on detailed analysis and research from Tables 6 and 8, and Appendix B.

	Mitigation Domain	Assets Affected	Threats Addressed	Mitigation Strategies
1.	Sensor Fusion techniques	LiDAR, RADAR	Jamming (DoS), Spoofing, Tampering	Implement Sensor Fusion techniques.
2.	Redundancy and Randomization	GNSS, LiDAR, Camera	Spoofing, DoS	Use Laser Pulses Strategically, Introduce Randomness in operation, Employ Hardware Redundancy Mechanisms, Enhanced Surveillance with Differential GNSS.
3.	Advanced Cryptographic Measures	ECUs, Telematic, Service laptop, OBD, Backoffice System, Bluetooth, Key fob or key less entry, vehicle's Uconnect system, LiDAR,	Spoofing, Tampering, Repudiation, Information Disclosure, Close Proximity Vulnerabilities,	Encrypt Communications and ensuring unpredictable password, Use Digital Signatures, Employ Cryptographic Checksum, Use Dedicated and Tamper-Resistant Hardware Security Modules, Secure A Key Management Process, Side-channel authentication, Encrypting the GPS code and implementing navigation message authentication (NMA) to embed signatures in satellite signals
4.	Network Security and Protocols	CAN Bus, External and In- vehicle Communication	Impersonation, Replay (Spoofing), DoS, Exploits, Jamming	Use LiBra-CAN,TrustNet, CAN ID Shuffling, SecOC, TLS, SRTP, Secure Communication Architecture (IDS And Firewall), Use Blockchain Technology, Timestamp all messages (to protect Replay Attack), Implement Frequency Agility, Packet Filtering.
5.	Physical and Operational Safeguards	Camera, LiDAR	Tampering	Employ Removable filters, Integrate Photochromic Lenses, Coatings are required for camera, LiDAR lenses, and ECU hardware, Seals and Physical locks or secure mounting systems of camera, Physical security in HSMs against tempering.
6.	Software and Firmware Integrity	Backoffice system, Telematics, CSM Sensors, VCM (software & firmware updates)	Flashing, Tampering, Hijacking	Implement Secure Boot, Provide Robust Authentication Method such as Digital Signatures, Implement Access Control for OEM Back-End Servers, Implement a robust and secure update mechanism for vehicle firmware and software.

Table 12 Mitigation Strategies

Table 12 presents a high-level overview of cybersecurity mitigation strategies specifically tailored for AVs. The researchers refined version categorizes the strategies into distinct mitigation domain, each addressing various aspects of AV cybersecurity.

1) Sensor Fusion techniques

This domain highlights to enhance the reliability and security of sensory data, which is crucial for the decision-making processes of AVs. The strategy of sensor fusion techniques such as Kalman Filters, Particle Filters and Neural Networks involves integrating data from multiple sensors to create a more accurate and resilient understanding of the vehicle's environment. This integration not only mitigates risks associated with data spoofing and jamming but also provides essential cross-verification capabilities among different sensor outputs.

2) Redundancy and Randomization

Building resilience and unpredictability into AV systems reduces the feasibility of successful targeted attacks. This domain includes implementing Redundancy by deploying multiple instances of critical sensors, such as GNSS, LiDAR, and cameras, ensuring that the failure or compromise of one sensor does not impact the overall functionality of the vehicle. Additionally, Randomization plays a crucial role by varying the operational parameters of these sensors. For example, altering the emission patterns of LiDAR complicates the efforts of attackers trying to predict or manipulate sensor behaviors. Moreover, Differential GNSS enhances the accuracy of standard GNSS systems by correcting positional inaccuracies, enabling precise location detection and the identification of even minor positional errors. This strategy effectively guards against threats like spoofing and DoS attacks, enhancing the security and robustness of AV systems.

3) Advanced Cryptographic Measures

This domain focuses on secure communication channels and ensures data integrity within the critical systems of AVs. The domain uses Encryption and Digital Signatures across various modules, including ECUs. Besides, the domain focuses on enhancing Wi-Fi hotspot encryption, implementing unpredictable password generation and side-channel authentication using AES (Advanced Encryption Standard) and electromagnetic radiation to secure communication of LiDAR. Specifically, for GPS data, advanced cryptographic techniques such as encrypting the GPS L1 P(Y) code and implementing Navigation Message Authentication (NMA) are used to embed cryptographic signatures in satellite signals. These methods protect GPS signals from spoofing and manipulation by malicious actors. The strategy encrypts data and verifies its integrity and origin, thereby guaranteeing that the data remains untampered and originates from a trusted source. Cryptographic Checksums are employed to validate the completeness and correctness of the data post-transmission, effectively preventing data tampering and corruption. Both strategies are used to mitigate close proximity vulnerabilities in Bluetooth, TPMS, Key Fob and Keyless Entry.

Furthermore, implement a robust key management process to enhance security. This strategy involves the secure generation, provisioning, usage, storage, and deletion/revocation of cryptographic keys. Inadequate key management can create exploitable vulnerabilities, especially in devices without direct user interfaces like embedded devices, which depend on remote administration and do not permit user-managed tasks like certificate management. Additionally, hardware-based cryptographic solutions can prevent the incorrect implementation of algorithms and reduce the occurrence of multiple software versions, offering more efficient resource usage. It is crucial to ensure the quality of hardware-based cryptography, as flaws in hardware implementations can affect all associated software.

4) Network Security and Protocols

This domain points to protecting AVs from network-based vulnerabilities and attacks, ensuring secure communication within the vehicle and with external entities such as other vehicles and infrastructure systems. Secure Communication Protocols such as LiBrA-CAN, TrustNet, CAN ID, SecOC, TLS and SRTP are employed to enhance in-vehicle communications, while Blockchain technology is utilized to facilitate secure, decentralized communications between vehicles and infrastructure. Filtering

communications in various network layers like ECU and sensor networks, VANET, and 2-5G communications involves nodes transmitting packets that are screened based on predefined rules and policies. Packets that match the criteria are either accepted or denied. Including a timestamp in messages, derived from a universal time source like UTC or GNSS, helps determine message recency and validity and prevents replay attacks. Employing a precision timing source, such as a hardware clock or Network time protocol (NTP), can enhance time accuracy.

Moreover, there are various attacks performed at radio frequencies. Implementing frequency agility in radio transmissions can significantly enhance resistance to jamming. By frequently changing the broadcast frequency within a defined band based on a pseudo-random algorithm, it becomes more challenging to disrupt the signal. Additionally, Intrusion Detection Systems (IDS) and Firewalls are integral components of the network security framework, actively monitoring network traffic for suspicious activities and effectively blocking unauthorized access and potential attacks.

5) Physical and Operational Safeguards

To safeguard the physical components and operational capabilities of autonomous vehicles, it is crucial to maintain the functionality and integrity of essential systems, including cameras and LiDAR. This domain ensures that these critical components remain uncompromised and continue to perform reliably in various environments. Camera Protection employs advanced techniques to shield onboard cameras from physical tampering and external manipulations. The integration of photochromic lenses enables cameras to automatically adjust to varying light conditions, reducing the risk of blinding by intense light sources. Removable filters can be employed to protect lens surfaces and improve image quality under different environmental conditions.

To enhance physical security further, special coatings are applied to camera and LiDAR lenses to prevent scratches, dust, and glare, maintaining visibility in diverse weather conditions. Seals around the camera housing keep out water and contaminants, ensuring consistent functionality. Physical locks or secure mounting systems are also utilized to guard against unauthorized removal or tampering, enhancing the overall security and resilience of these critical sensory systems. Similarly, protective coatings on ECU hardware serve to shield components from physical manipulation and environmental damage, while also enhancing overall durability. These coatings can be made from various materials such as conformal coatings, potting compounds, or ruggedized materials designed to withstand harsh conditions. According to ISO 26262, which outlines the Automotive Safety Integrity Level (ASIL) for functional and safety verification of automotive semiconductors, achieving high reliability and stability in electronic components is crucial. Protective coatings contribute significantly to meeting these stringent safety standards, ensuring that ECUs perform reliably under various environmental stressors.

Furthermore, HSMs are dedicated hardware devices designed to provide secure cryptographic key management and operations as mentioned before. In addition to their cryptographic capabilities, HSMs include physical security measures to protect against tampering and environmental threats. These features can include tamper-evident seals, intrusion detection systems, and hardened enclosures.

6) Software and Firmware Integrity

To maintain the integrity and security of the software and firmware within AV systems in the update process, it is essential to implement robust protections against unauthorized modifications, exploitation, and hijacking. Secure boot mechanisms essentially consist of signing the software with a private key to ensure that the vehicle’s software starts up in a verified and secure state. This process is critical for preventing the execution of malicious code at system initialization. Additionally, robust authentication methods such as digital signature play a vital role during software updates. These methods verify the authenticity of update packages and are crucial for preventing the installation of unauthorized or corrupted updates and avoiding rollback to vulnerable firmware versions. Furthermore, Access controls for OEM back-end servers and recovery protocols, such as reverting to a previous version if the OTA update fails to boot, should be implemented.

Additionally, the importance of implementing a robust and secure update mechanism for vehicle firmware and software is emphasized. This process involves secure delivery and rigorous verification of updates to prevent malicious modifications. It ensures that only authorized and validated updates are applied, safeguarding the system against potential security threats.

5.3.2 Mitigation strategies for evolving threats of AVs

Based on the findings in section 5.1.4, AVs are susceptible to specific threats, including signal interference and manipulation, physical manipulation and deception, as well as unauthorized access and control. Implementing targeted mitigation strategies is crucial to effectively counter these threats and enhance AV security. The following in Figure 7 outlines how mitigation strategies can be implemented.

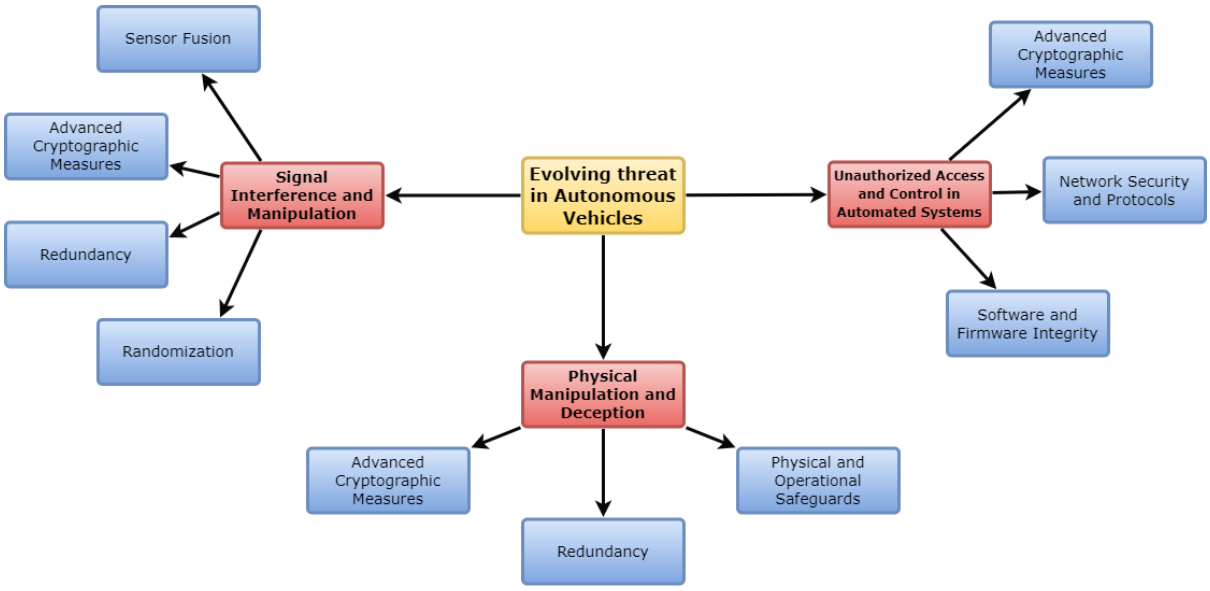

Figure 7. Mitigation Strategies for evolving threat in AVs

To counter threats of **signal interference and manipulation** in AVs, implementing sensor fusion techniques, advanced cryptographic measures, redundancy, and randomization is essential. Sensor fusion effectively integrates data from diverse sources, enhancing the reliability of decisions by providing alternative validation points, which are crucial when signals are spoofed or jammed. Ensuring that data transmitted between sensors and processing units remains secure is vital. Advanced cryptographic techniques, such as encrypting GPS L1 P(Y) codes and implementing Navigation Message Authentication (NMA), add cryptographic signatures in satellite signals. These techniques prevent the signals from potential interception or alteration. Additionally, utilizing side-channel authentication methods based on electromagnetic radiation ensures secure data communication from sensors like LiDAR. Redundancy guarantees continuity of operations, allowing other systems to take over if a critical sensor is compromised. Lastly, randomization of operational parameters disrupts predictable patterns, thwarting exploitations based on these vulnerabilities.

At the same time, addressing threats from **physical manipulation and deception** relies on a redundancy strategy that enables cross-verification among different sensors. This approach ensures that even if one sensor is compromised, the overall functionality of the vehicle remains unaffected. In this case, advanced cryptographic measures also play a key role in safeguarding the communication of sensor data to the processing unit. For instance, side-channel authentication for LiDAR uses a cryptographic device in a vehicle performing AES calculations, to ensure the authenticity and integrity of the data. Furthermore, implementing physical and operational safeguards is vital for protecting sensors from tampering, ensuring their operational integrity, and maintaining the overall security posture of the vehicle.

Mitigating **unauthorized access and control** in automated systems requires advanced cryptographic measures. Utilizing strong cryptographic protocols like AES, a robust encryption standard, ensures that data transmitted within the system remains confidential and cannot be read by unauthorized entities. For example, encrypting GPS codes with AES prevents attackers from intercepting and manipulating positioning data. Robust key management is essential, as cryptographic keys must be securely stored to prevent unauthorized access. HSMs provide a secure storage solution, protecting keys from physical and logical attacks. Network security is bolstered by secure communication protocols such as LiBrA-CAN, TrustNet, CAN ID, SecOC, TLS, and SRTP, along with intrusion detection systems and firewalls, which are instrumental in thwarting unauthorized access and safeguarding control systems. Additionally, software and firmware integrity are maintained through secure boot processes and rigorous authentication during updates, preventing the installation of malicious or unauthorized software, thus ensuring the system's resilience against cyber threats.

6 Contribution

This chapter presents two contributions: the development of an ontology for understanding cybersecurity threats faced by AVs and the integration in STRIDE threat model.

6.1 Ontology

The research contributes to the field by developing an ontology for understanding the cybersecurity threats faced by AVs, focusing on assets, threats, and mitigation strategies. The ontology serves as a knowledge model, capturing domain-specific concepts, relationships, and entities related to autonomous vehicle cybersecurity.

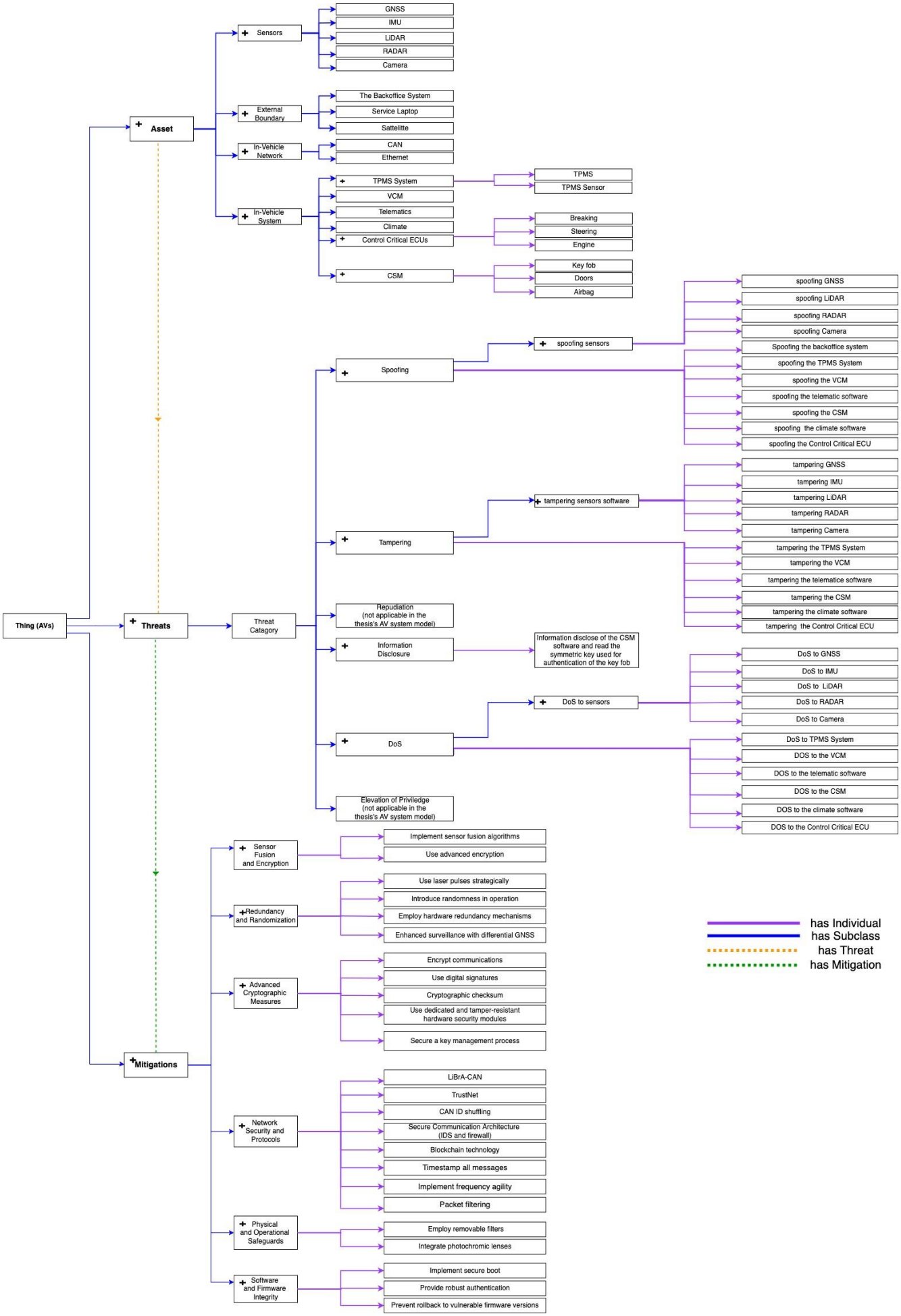
Thing (AVs)
Asset
Sensors
GNSS
IMU
LiDAR
RADAR
Camera
External Boundary
The Backoffice System
Service Laptop
Sattelitte
In-Vehicle Network
CAN
Ethernet
In-Vehicle System
TPMS System
TPMS
TPMS Sensor
VCM
Telematics
Climate
Control Critical ECUs
Breaking
Steering
Engine
CSM
Key fob
Doors
Airbag
Threats
Threat Catagory
Spoofing
spoofing sensors
spoofing GNSS
spoofing LiDAR
spoofing RADAR
spoofing Camera
Spoofing the backoffice system
spoofing the TPMS System
spoofing the VCM
spoofing the telematic software
spoofing the CSM
spoofing the climate software
spoofing the Control Critical ECU
Tampering
tampering sensors software
tampering GNSS
tampering IMU
tampering LiDAR
tampering RADAR
tampering Camera
tampering the TPMS System
tampering the VCM
tampering the telematice software
tampering the CSM
tampering the climate software
tampering the Control Critical ECU
Repudiation (not applicable in the thesis's AV system model)
Information Disclosure
Information disclose of the CSM software and read the symmetric key used for authentication of the key fob
DoS
DoS to sensors
DoS to GNSS
DoS to IMU
DoS to LiDAR
DoS to RADAR
DoS to Camera
DoS to TPMS System
DOS to the VCM
DOS to the telematic software
DOS to the CSM
DOS to the climate software
DOS to the Control Critical ECU
Elevation of Priviledge (not applicable in the thesis's AV system model)
Mitigations
Sensor Fusion and Encryption
Implement sensor fusion algorithms
Use advanced encryption
Redundancy and Randomization
Use laser pulses strategically
Introduce randomness in operation
Employ hardware redundancy mechanisms
Enhanced surveillance with differential GNSS
Advanced Cryptographic Measures
Encrypt communications
Use digital signatures
Cryptographic checksum
Use dedicated and tamper-resistant hardware security modules
Secure a key management process
Network Security and Protocols
LiBrA-CAN
TrustNet
CAN ID shuffling
Secure Communication Architecture (IDS and firewall)
Blockchain technology
Timestamp all messages
Implement frequency agility
Packet filtering
Physical and Operational Safeguards
Employ removable filters
Integrate photochromic lenses
Software and Firmware Integrity
Implement secure boot
Provide robust authentication
Prevent rollback to vulnerable firmware versions
has Individual
has Subclass
has Threat
has Mitigation

Figure 8. Overview of cybersecurity ontology in AVs

Figure 8 presents a general view of the proposed ontology. The ontology consists of three main classes: Asset, Threat, and Mitigation.

1) **Asset Class**: At the core of the ontology is the Asset class, which encompasses the critical components and systems within AVs that are susceptible to cybersecurity threats. This class is further delineated into four sub-classes: Sensors, External Boundary, In-vehicle network, and In-vehicle system. By categorizing assets based on their functional roles and significance within the AV ecosystem, the ontology provides a comprehensive inventory of the components requiring protection against cyber threats.
2) **Threat Class**: Complementing the Asset class, the Threat class categorizes cybersecurity threats facing AVs, providing a systematic framework for threat analysis and classification. Within this class, a sub-class named Threat Category is defined, organizing threats according to the STRIDE threat modeling methodology. Additionally, sub-classes of Threat Category for specific sensors asset; Spoofing, Tampering, and DoS, are defined to highlight threats targeting individual sensor components. Some threats categories, such as repudiation and elevation of privilege, are deemed not applicable for threat analysis and risk assessment in this research context.
3) **Mitigation Class**: In response to identified threats, the Mitigation class offers a categories of mitigation strategies aimed at mitigating cybersecurity risks and enhancing the resilience of AV systems. This class encompasses six sub-classes: Sensor Fusion and Encryption, Redundancy and Randomization, Advanced Cryptographic Measures, Network Security and Protocols, Physical and Operational Safeguards, and Software and Firmware Integrity. Each sub-class represents a distinct category of mitigation techniques tailored to address specific vulnerabilities and safeguard assets against potential threats.

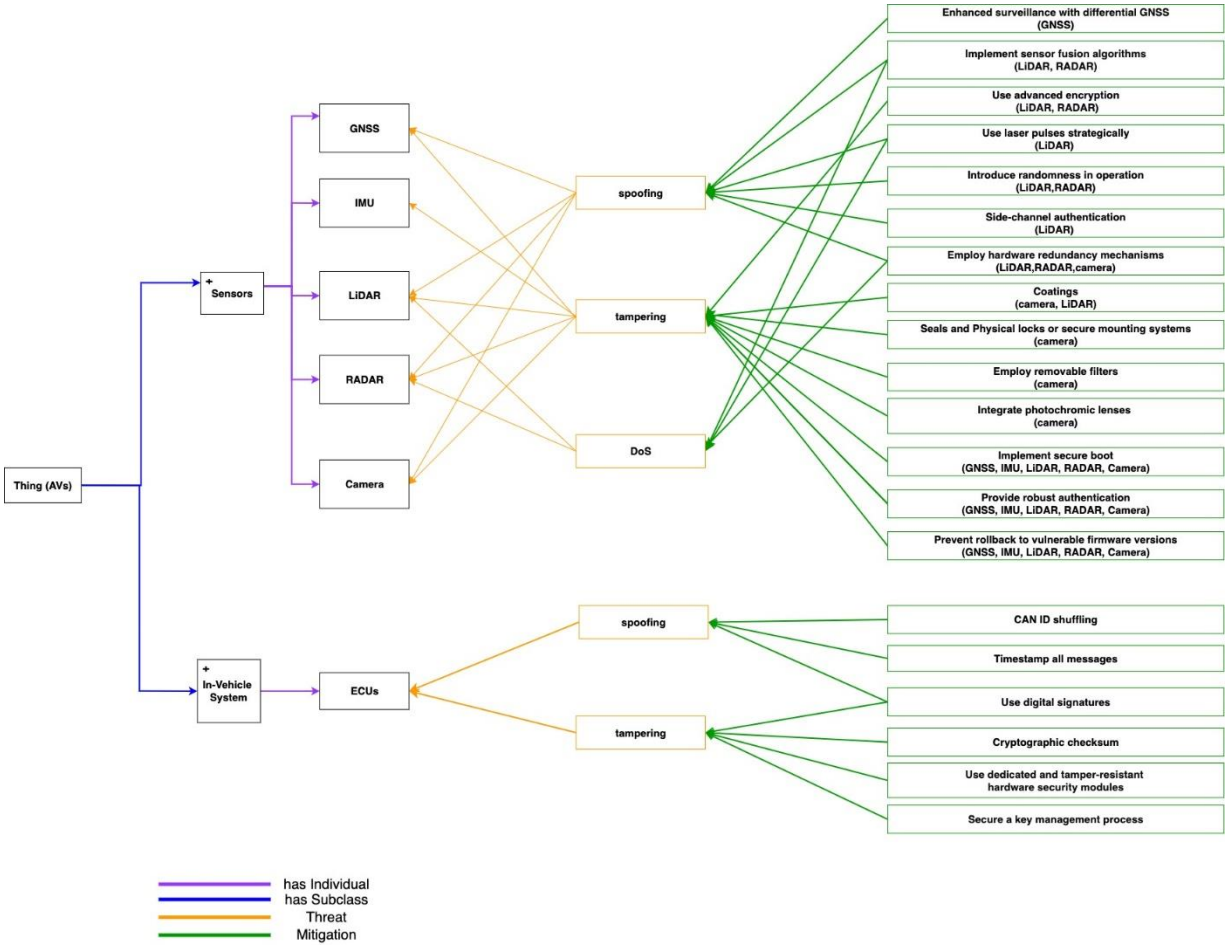

Figure 9. Visualization the relation of asset, threat, mitigation of sensors and in-vehicle system

In Figure 9, the ontology illustrates the relationships between assets, threats, and mitigation strategies within the unique system architecture of AVs. This diagram highlights the evolving threats identified in the results of research question 1, as well as the corresponding mitigation strategies identified in the results of research question 3.

At the core of this ontology are the subclasses of sensors: GNSS, IMU, LiDAR, RADAR, and camera. Each sensor subclass is associated with specific cybersecurity threats and corresponding mitigation measures tailored to address these risks effectively. For instance, GNSS predominant threat identified is spoofing, where malicious actors attempt to deceive the system by transmitting false signals. Spoofing attacks can lead AVs to receive inaccurate positioning information, disrupting navigation and potentially causing safety hazards. To mitigate this threat, the strategy involves enhanced surveillance with differential GNSS techniques employed. These techniques involve sophisticated monitoring and validation mechanisms to differentiate between authentic and spoofed signals, thereby enhancing the integrity and reliability of GNSS data used by AVs for navigation purposes.

One of the threats facing AVs is spoofing ECUs, whereby attackers can disrupt the normal operation of the AV in various ways. For example, they may send fake communication to different ECUs that misleading vehicle's critical control system. Robust mitigation strategies are imperative to counteract these threats. For spoofing attacks, mitigation measures include techniques such as CAN ID shuffling, timestamping all messages, and using digital signatures to enhance the authenticity and integrity of data exchanged between ECUs, thereby mitigating the risk of spoofing attacks.

Another significant threat facing AVs is tampering with ECUs. Attackers may gain physical or remote access to the ECUs in an AV. Once access is obtained, they can manipulate the software or firmware of the ECUs to compromise critical control systems within the vehicle. To effectively mitigate this

threat of tampering, a multifaceted approach is adopted. This includes encrypting communications to prevent unauthorized access, utilizing digital signatures to verify the authenticity of commands and data transmitted, implementing cryptographic checksums to detect tampering or unauthorized alterations to data, deploying dedicated and tamper-resistant hardware security modules to protect sensitive information and functions, and establishing a secure key management process to ensure the confidentiality and integrity of cryptographic keys used for authentication and encryption purposes.

6.2 Integration in STRIDE

An additional contribution from this thesis involves the integration of “spoofing” from the STRIDE threat model with “environment”, enriching the TARA methodology by capturing AV threats comprehensively.

This integrated approach ensures a more comprehensive assessment of AV threats by considering how spoofing attacks exploit vulnerabilities in the vehicle’s perception systems, particularly sensors influenced by surrounding objects and signals. By incorporating spoofing within the STRIDE model with environmental factors, stakeholders can gain a more nuanced understanding of potential threats, covering both cyber and physical domains. Such an approach enables the identification of threats arising from various environmental conditions, ensuring that the entire spectrum of security risks is adequately addressed.

Considering environmental factors in conjunction with spoofing enriches threat modeling with realistic scenarios encountered by AVs. Adversaries may exploit environmental conditions, such as obstructed sensor views or manipulated signals, to deceive vehicle sensors and compromise their perception capabilities. By integrating spoofing within the STRIDE model with environmental considerations, TARA can effectively simulate and assess these real-world threat scenarios. This capability enables more accurate threat identification and the development of risk mitigation strategies tailored to diverse environmental settings, thereby enhancing the resilience of autonomous vehicles against spoofing attacks.

Integrating spoofing within the STRIDE model with environmental considerations also raised awareness of the multifaceted nature of AV threats among stakeholders during the analysis of this study. By illustrating how spoofing attacks exploit both technical vulnerabilities and environmental conditions, this integrated approach underscores the importance of non-traditional security measures in safeguarding autonomous vehicles against evolving threats.

7 Conclusions and Discussion

7.1 Discussion

7.1.1 Result Discussion and Relation to Previous Research

The result findings presented from document analysis and the TARA offer significant insight into cybersecurity in AVs.

The findings of research question 1 highlight three primary threats to AVs. Among these threats, signal interference and sensor manipulation stand out as significant concerns. AVs heavily rely on external signals and connectivity, making any tampering with these signals a potential threat to their ability to accurately perceive their surroundings. Additionally, threats related to physical manipulation and deception pose challenges due to the advanced sensor technologies used in AVs. Moreover, unauthorized access and control present risks to the integrity of automated control systems.

It is evident that the threats targeting AV sensors are particularly significant, as they directly impact the vehicle's ability to navigate safely and effectively. Sensors play a pivotal role in enabling AVs to perceive and interact with their environment, making them essential for safe and efficient operation. This finding is consistent with previous research efforts conducted by Giannaros et al. [6], which emphasized the vulnerabilities within sensor systems as fundamental elements for AV functionality. These vulnerabilities have been pinpointed as potential targets for cyber-attacks, posing threats to the safety and reliability of AVs.

Research question 2 delves into evaluating the limitations and strengths of the TARA methodology under the framework of the ISO/SAE 21434 when applied to address the cybersecurity threats in autonomous vehicles. The findings underscore limitations in utilizing the STRIDE threat modeling approach to analyze specific threats targeting AV sensors and environments, as well as challenges related to expertise in feasibility analysis when dealing with non- traditional cyber security threats to sensors.

The encountered limitation during the risk assessment stage, particularly when employing the STRIDE model to analyze the environmental threats, resonates with concerns previously noted in research. Sheik et al. [13] emphasizing the inadequacy of conventional threat modeling techniques in comprehensively capturing the nuanced and dynamic threat environments inherent to AVs. While the specific points of critique may differ, both this current study and the aforementioned research converge on the consensus that the reliance of the Threat Agent Risk Assessment (TARA) methodology on the STRIDE model introduces constraints that obstruct its effectiveness in addressing emerging cybersecurity challenges in AVs.

Moreover, the findings addressed the partially subjectivity and uncertainty associated with TARA Feasibility ratings when assessing evolving non-traditional cybersecurity threats discovered in AVs. This perspective resonates with similar viewpoints regarding expert judgment presented in Khalid's study [14], which emphasizes TARA's heavy reliance on security experts while highlighting the importance of considering damage scenario categories and impact ratings. Similarly, Benyahya et al. [8] contribute to this discourse by cautioning against potential biases of relying on expert judgment. They argue that estimations based on expert evaluations may lead to overconfident or underconfident results, thereby introducing bias into risk computation processes.

Apart of the findings from evaluating the TARA methodology, the researchers recognized that the ISO/SAE 21434 standard allows for the consideration of additional impact categories beyond the predefined ones, which include operational, safety, privacy, and financial considerations. The researchers suggest extending these categories to include "reputation damage" because cyber security incidents can have far-reaching implications for public perception and trust in AV automotive manufacturers [48],[51]. People attribute greater importance to fatalities involving AVs compared to those caused by human drivers in conventional vehicles (CVs), especially if AV fatalities result from

deliberate misuse, such as exploiting AV software vulnerabilities [87]. Additionally, media coverage of damage resulting from accidents involving AVs such as demonstrations of remote hacking and manipulation, has gained significant attention. In non-AV scenarios, if an accident occurs, it is commonly understood that the driver is responsible. However, the research [88] indicates that the AV manufacturer was selected by 70.7% of respondents as the entity that should accept the highest level of liability for AVs in case of accidents, suggest that the public perceives trust in safe, reliable, and secure modes of AV transportation to be linked to the AV manufacturer. Failure to address these concerns could undermine consumer trust in AV automotive manufacturers, potentially leading to decreased consumer adoption and irreparable harm to the brand's image, affecting its long-term viability and competitiveness in the market.

The findings of Research Question 3 present a set of high-level cybersecurity mitigation strategies tailored specifically for the unique threats identified in AVs. To counter threats such as signal interference and manipulation, implementing sensor fusion techniques, advanced cryptographic measures, redundancy, and randomization are essential. Similarly, addressing risks stemming from physical manipulation and deception necessitates the deployment of redundancy, advanced cryptographic techniques, and the implementation of physical and operational safeguards to protect sensors from tampering. Moreover, mitigating unauthorized access and control in automated systems demands the application of advanced cryptographic measures, network security protocols, and strategies to ensure the integrity of software and firmware.

In the automotive industry, UN R155 outlines standards and requirements for ensuring cybersecurity in vehicles, emphasizing comprehensive risk management practices across the entire lifecycle and throughout the supply chain. Although it emphasizes broader management practices and system-level security, this research observed that UN R155 does not explicitly address component-level protections for elements such as sensors and cameras.

Addressing this gap, this research outlines specific strategies to enhance the integrity and confidentiality of sensor and camera data in AVs by advocates for sensor fusion techniques to ensure data reliability and accuracy, underscoring the importance of sensor security as these components are crucial for an AV's interaction with its environment and overall vehicle safety. Additionally, it includes physical protection measures for cameras to guard against tampering and maintain operational reliability under various environmental conditions. These physical security measures, not thoroughly covered by UN R155, are critical as they complement the digital cybersecurity strategies by providing a holistic approach to protecting AVs from both digital and physical threats.

7.1.2 Limitations in Study Design

This research's reliance on document analysis and the TARA methodology guideline offers valuable insights into AV cybersecurity threats. However, challenges related to data availability and accessibility constrain the depth and accuracy of the analysis. The study encounters limitations stemming from the confidentiality of AV information. Manufacturers often refrain from publishing detailed information on AV cybersecurity due to concerns about proprietary technology, competitive advantage, and potential security risks. As a result, access to comprehensive data sets and insights into AV cybersecurity practices is restricted, impeding the thorough assessment of vulnerabilities and risk factors. This confidentiality barrier underscores the challenge of obtaining accurate and in-depth information necessary for a comprehensive analysis of AV cybersecurity threats.

7.1.3 Ethical and Societal Consequences

The absence of a human driver in AVs significantly amplifies the ethical and societal consequences, particularly in the realm of cybersecurity. The findings of this study underscore the vulnerabilities inherent in AV technology, presenting a threat that extends far beyond traditional vehicle safety concerns. One of the most pressing ethical dilemmas arises from the potential for malicious actors to exploit the lack of human oversight in AVs. Without a human driver to intervene or override automated

systems, these vehicles become susceptible to hacking, tampering, and remote control by unauthorized individuals. Such breaches could have catastrophic consequences, ranging from minor disruptions to traffic flow to large-scale accidents resulting in injury or loss of life. Moreover, the deployment of AVs raises profound questions about liability and accountability in the event of accidents or system failures. With no human driver to assume responsibility, determining fault and allocating liability becomes increasingly complex, potentially leading to legal and ethical ambiguities.

7.1.4 Utilization of IT Tools

The thesis employs various IT tools to enhance the writing process, including Grammarly, Quillbot, and ChatGPT. Grammarly serves as a valuable tool for grammar and spelling checks, ensuring the clarity and coherence of the written content. Its advanced algorithms assist in identifying and rectifying grammatical errors, punctuation mistakes, and stylistic inconsistencies, thereby improving the overall quality of the thesis. Additionally, Quillbot aids in paraphrasing and rephrasing sentences, offering alternative wordings and enhancing the fluidity of the text. ChatGPT, with its language processing capabilities, further contributes to the writing process by providing grammar correction suggestions and enhancing the overall coherence and structure of the thesis.

7.2 Limitation of the study

While the study provides insights into the evolving threat landscape and mitigation strategies for AV cybersecurity, it may not cover all possible scenarios or emerging threats. The dynamic nature of cybersecurity threats requires ongoing monitoring and adaptation of strategies, which may not be fully captured in a single study. The rapid pace of technological innovation in the automotive industry means that newer AV models may incorporate advanced features or security measures do not present in older models. The study's findings may be more relevant to AVs with similar technological specifications or those developed around the time of the study.

Additionally, the findings of the study may not be universally applicable to all AV contexts. Different manufacturers may employ unique architectures, technologies, and cybersecurity protocols in their AVs. These variations can lead to differences in vulnerability profiles, threat landscapes, and mitigation strategies, limiting the generalizability of the study's findings to the broader AV industry. The manufacturers also guard proprietary information and trade secrets related to their AV technologies and cybersecurity measures. As a result, certain details or insights obtained during the study may not be publicly available.

7.3 Contribution

The research introduces an ontology tailored specifically to understand cybersecurity threats faced by AVs. This ontology provides an understanding of the unique components susceptible to cyber threats within the AV ecosystem. Through detailed subclassification, such as differentiating threats based on the STRIDE methodology and delineating mitigation strategies, the ontology serves as a knowledge model for analyzing and addressing cybersecurity challenges in AVs. Moreover, the research also integrating "spoofing" from the STRIDE threat model with "environment" thereby enriching the TARA methodology and allowing for a more comprehensive understanding of AV threats. By incorporating these contributions, stakeholders in the automotive industry can navigate the complex landscape of cybersecurity, thereby fostering the deployment of safer and more secure AVs.

7.4 Future Research

As the field of AVs and cybersecurity continues to evolve rapidly, there remain several aspects for future research that could further enhance understanding of the challenges and opportunities in this

domain. Future research can consider incorporating the TARA methodology with a wider range of components that contribute to the functionality and security of AVs. The components may include, communication networks, entertainment systems, and external interfaces such as cellular networks and V2X communication.

7.5 Conclusion

The research comprehensively explores the evolving threat landscape for AVs, which includes signal interference and manipulation in sensors as it heavily operates dependence on external signals and connectivity, as well as physical manipulation and deception, can compromise the vehicles' ability to accurately interpret their surroundings. Additionally, automated control systems elevate the risks associated with unauthorized access and control.

This research includes proposed mitigation strategies to address these evolving threats. For combating signal interference and manipulation, a combination of sensor fusion, advanced cryptographic techniques redundancy, and randomization emerges as a promising approach. Threats from physical manipulation and deception can be mitigated by implementing redundancy, physical and operational safeguards, and advanced cryptographic techniques to safeguard the communication of sensor data to the processing unit. Unauthorized access and control in automated systems can be addressed through advanced cryptographic measures, network security protocols, and software and firmware integrity strategies.

The evaluation of the TARA methodology within the ISO/SAE 21434 framework, revealed its limitations which are the use of STRIDE threat modeling to analyze specific threats to AV sensors and environments, as well as challenges related to expertise in feasibility analysis when dealing with non-traditional cyber security threats to sensors.

The thesis proposes two contributions. Firstly, it introduces an ontology that provides an understanding of the unique components in AVs susceptible to cyber threats, along with delineating mitigation strategies. Secondly, the consideration of integrating "spoofing" from the STRIDE threat model with "environment" to enriches the TARA methodology in comprehensively capturing AV threats.

In conclusion, the thesis provides a structured analysis of the current threat landscape, evaluates existing methodologies, and suggests enhancements to improve the resilience of AV systems against cyber threats, contributing valuable insights to the field of automotive cybersecurity.

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Appendix A – Threats in AVs in existing research

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
Sensors	GPS (Microwave)	Jamming	Employing stronger signals within the same frequency range takes over the signals from satellites, due to weak GPS signals from satellites, which have the long distances they travel.	The victim's GPS loses the lock of the signals.	[32]
		Spoofing (incorporates an initial phase of GPS jamming [6])	Mislead victims to incorrect positions (and times) by creating fake signals that diverge from their actual location. This involves first jamming the victim's GPS receiver to disrupt signal reception, followed by transmitting a stronger, deceptive signal to override the satellite signals.	The victim's GPS either stops receiving signals or changes unexpectedly.	[32]
		Meaconing	It involves capturing GPS signals and intentionally delaying their retransmission.	The action led to inaccuracies in the GPS receiver position calculations.	[6]
		Replay	It involves capturing GPS signals and broadcasting them again at a different time or place.	The action can deceive the GPS receiver into incorrectly determining its location or time.	[6]
	LiDAR (Infrared Laser)	Relay	An attacker uses a transceiver to intercept and delay the laser pulses emitted by the LiDAR, then sending altered signals back to it. This manipulation creates the illusion of non-existent obstacles at fixed positions.		[32]
		Inject of fake obstacles	By exploiting the LiDAR's rotational scanning method and the speed of light, the attacker can simulate even closer fake obstacles by redirecting the intercepted laser pulse from different angles e.g., a barrier in front of the vehicle [37]	Target vehicles might stop or divert upon detecting these false obstacles. If several vehicles encounter this illusion, it could lead to significant traffic jams or collisions as AVs attempt to evade these non-existent hazards [37].	[32], [37]
		Hidden objects	An attacker could employ materials that absorb light to remain undetected; if the object is coated with reflective materials, the vehicle might fail to detect it.	The result could be physical harm or potentially fatal injuries to the targeted vehicle.	[37]
		Jamming	It involves sending laser beams at the same frequency as the LiDAR uses, interfering with its scanning capabilities.	Potential rendering is unable to accurately detect real obstacles.	[32],[37]
		Electromagnetic interference (EMI)	LiDAR may be susceptible to intentional electromagnetic interference (IEMI), impacting the time-of-flight components integral to contemporary LiDAR systems.	Such interference can lead to the autonomous vehicle's perception system incorrectly detecting or categorizing objects, or seeing obstacles that are not there, posing a threat to road safety.	[6]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
	RADAR (Microwave)	Jamming	An attacker might employ noise to degrade the radar's signal. This compromised radar could malfunction, rendering the vehicle unable to perceive its surroundings.	Noise impacts several autonomous vehicles; it could disrupt traffic or even lead to accidents. While an attack does not directly result in data breaches, it could lead to physical harm.	[32],[37]
		Hidden Objects	Current advancements allow objects to evade radar detection, a technique already utilized in military aerospace. Aircraft or objects can conceal themselves either by altering their usual reflective shape or employing materials that absorb radar waves.	This type of threat does not lead to data breaches but could result in physical damage or direct harm to individuals.	[37]
		Fake objects	Attackers emit fake radar signals to initiate the assault. Subsequent vehicles recognize these deceptive signals and respond accordingly.	While the assault does not lead to the exposure of sensitive information, it could inflict physical harm to structures, such as collisions resulting from vehicles attempting to evade these fictitious objects.	[37]
	GNSS	Spoofing	Attackers might manipulate devices to misdirect the vehicle to incorrect locations or routes. GNSS spoofing, which misleads more than merely jamming GNSS signals, poses a greater threat. In the absence of GNSS, AVs would rely on alternatives like V2V communication or Simultaneous Localization and Mapping (SLAM) for navigation and collision avoidance.	If the information is inaccurate and goes unnoticed, AVs would rely on the incorrect GNSS data and make erroneous decisions, potentially resulting in accidents and serious injuries.	[37]
		Jamming	Attackers transmit signals of higher power, overpowering the inherently weak GNSS signals, causing the genuine signals to be disregarded. Detecting such jamming attacks is challenging due to potential signal degradation from interference or limited satellite visibility.	Without GNSS signals, autonomous vehicles struggle with navigation and positioning, but V2V communication can serve as an alternative for coordinated navigation.	[37]
	Camera	Blinding	Attackers using lasers or LED arrays can disrupt cameras by rapidly toggling the light to confuse them making the vehicle may be more vulnerable to unperceived obstacles [8]. Additionally, shining a laser directly at a vehicle's camera from a very close distance (less than half a meter) for even a short moment can cause permanent damage, thereby interrupting autonomous functions [32].	The attack is unlikely to lead to severe injuries since it can be easily detected, and AVs utilize data from multiple sensors. When the cameras fail, other sensors are still capable of perceiving the environment. [37]	[6], [32],[37]
		Mislead camera (fake images)	By remotely manipulating cameras, attackers can introduce false imagery to deceive them.	The tactic is riskier than blinding the cameras, as detecting such manipulations is more challenging. While a system or user can readily notice vision impairment, identifying misleading imagery takes longer, increasing the potential for harm and elevating the threat level to a high.	[37]
		Tampering	Modify the camera in an ACN (Automatic Crash Notification) system to skew crash data accuracy. Altering the port connected	Possibly affect misguided emergency responses, delayed assistance, and complications in	[35]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
			to the ACN camera can undermine the reliability of data transmission.	processing insurance claims. Such tampering jeopardizes lives, impedes accountability, and diminishes the system's efficacy.	
		Denial of Service	Attackers disrupting the camera can interfere with crash data gathering and transmission, resulting in delayed or unreported crash notifications. Similarly, DoS attacks on the port connecting the ACN (Automatic Crash Notification) camera can halt data transmission.	The result leads to compromised emergency responses, delaying notifications, complicating insurance claim procedures, and reducing the ACN system's overall efficiency.	[35]
		Elevation-of-privilege	Unauthorized access to ACN camera systems can lead to higher-level control breaches, jeopardizing the accuracy of accident data and the functioning of the system.	The result is that manipulated crash reports, postponed emergency assistance, and obstacles in handling insurance claims, diminishing the efficacy of ACN systems overall.	[35]
		Repudiation	Malicious users disclaim their participation in documented incidents, causing legal and accountability complications.	The actions compromise the dependability of accident data, emergency responses, and insurance claims processing, eroding confidence in ACN systems.	[35]
	Ultrasonic (Sound wave)	Jamming	Attackers reduce the Signal-to-Noise Ratio (SNR) of ultrasonic sensors in vehicles (tested in Audi, Volkswagen, Tesla, and Ford) by emitting continuous ultrasound. Another experiment shows that jamming attacks effectively disrupt Tesla's self-parking and summon modes, causing the vehicles to collide with obstacles unnoticed [32].	The vehicle's failure to warn drivers about obstacles.	[24], [32]
		Spoofing	An attacker employs precisely engineered ultrasound to simulate fake obstacles, generating phantom barriers where none exist within the sensor's detection range. This technique can conjure virtual obstacles for commercial sensors and sensors built into AVs.	The attack can readily lead to confusion in the autonomous vehicle's decision-making processes, especially when multiple obstacles are introduced.	[32]
Cab Security Module	Key Fob/ Keyless entry	Jamming	When the user presses the "close" button, an attacker can jam the locking signal with interference, leaving the user unaware that the door stays unlocked. This enables break-ins, utilizing intentional electromagnetic interference (IEMI) without needing cryptographic or chip analysis, making the attack simple and cost-effective.	Attackers can access the vehicle.	[6], [32], [38]
		Replay	Attackers can intercept and record the signal exchange between a key transponder and the vehicle's receiver, using the recording to unlock an unsupervised vehicle. However, modern vehicles often use rolling codes, making this attack less effective. Attackers might combine this with other methods, such as jamming to	Attackers can access the vehicle.	[32]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
			capture a valid “close” signal to replay after a break-in, ensuring the vehicle locks correctly afterward. They may continue to intercept, jam, and record signals until capturing the desired one.		
		Relay	Attackers use devices to bridge the distance between a vehicle’s key fob and the vehicle itself. The attack bypasses the system’s distance limitations, tricking the car into thinking the key fob is closer than it actually is. The method exploits the passive keyless entry and start (PKES) system, where the vehicle unlocks when the key is nearby and starts when the key is detected inside. For example, Attackers use two antennas: one near the car to relay signals to the car’s receiver, and another following the key fob owner to relay signals from the key. This can effectively make the car and key fob communicate over vast distances (up to 3000 km in tests), without the need to crack cryptographic authentication.	Allow for unauthorized access and engine start, making parked vehicles vulnerable to theft even when the owner is far away.	[32]
		Cryptographic Analysis Attacks	Attacks on vehicle systems target both physical-layer communications and higher-layer cryptographic algorithms. Early remote keyless entry systems, using fixed codes without encryption, were vulnerable to replay attacks. Rolling codes introduced for authentication improved security, but systems like Keeloq [16] remain susceptible to cryptographic and side-channel attacks. Attackers can also exploit entry system PCBs to analyse and clone firmware, particularly in systems using a common master key, allowing unauthorized access to vehicles. Studies have demonstrated the feasibility of cloning VW group remotes and decrypting signals by analyzing cryptographic schemes, with techniques such as correlation attacks quickly compromising keys.	Enable unauthorized engine start.	[32]
	Immobilization	Cryptanalytic and fault injection	Vehicle immobilizer attacks exploit weaknesses in the electronic system that prevent the engine starting without the correct key fob. Vulnerabilities in transponder schemes like Hitag2 and Megamos arise from flawed cipher designs and poor pseudorandom number generation, allowing attackers to bypass security and access private key information quickly.	Attackers can access and start the vehicle.	[32]
	Voice Controllable Systems (VCSs)	Hidden voice command, Dolphin Attack and LipRead	DolphinAttack transmits inaudible voice commands to devices like Amazon Echo from up to 5 ft away by exploiting microphone nonlinearity. LipRead extends this technique to 8 meters using multiple speakers to keep signals inaudible. Both require special ultrasound-emitting devices, making the attacker potentially detectable and limiting the practicality of these hidden command attacks.	Unauthorized control over speech recognition systems, potentially leads to privacy breaches, security vulnerabilities, and unauthorized actions or access.	[32], [34]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
In-vehicle Systems (Control Critical ECUs, TPMS)	ECUs	Tampering	Unauthorized access to the ECU can lead to tampered functionality, triggering false crash notifications or altering insurance log files.	The actions can cause unwarranted emergency responses, resource waste, confusion, financial losses for insurers, higher premiums for users, and breaches of vehicle owner privacy through the collection and exposure of sensitive data.	[35]
			Tampering with the connection between the ECU and TCU	The action can cause delays or failures in crash data transmission, leading to slower emergency response times when immediate assistance is vital. Furthermore, it can cause service outages or system failures, completely disabling ACN functionality and preventing any crash-related notifications	[35]
		Denial of Service	The overload of requests or the processing of malformed packets can exhaust the ECU's resources, leading to a slowdown or complete halt in its operation.	The ECU could shut down the entire automatic crash notification system, preventing it from receiving accident data from sensors and communicating with the components that activate the alarm	[35]
		Elevation of Privilege	Gaining access to the ECU usually permits only data viewing, but an elevation-of-privilege attack enables attackers to modify data and harm the ECU or connected devices. Motivations for these attacks come from erasing diagnostic codes and altering configurations to activating premium features or inflicting irreversible electronic damage, including ransomware attacks.	Enable attackers to modify data and harm the ECU or connected devices.	[35]
		Repudiation	Without a solid non-repudiation mechanism, any harmful actions on the ECU—and by extension, the whole vehicle—can later be denied.	The lack of accountability could lead to significant operational and safety risks, compromising vital functions like the ACN.	[35]
		Software Flashing	The attacks target the embedded systems in vehicles, like those controlling gear shifts, steering, and climate controls. These systems are reprogrammable, allowing updates for bug fixes and new features without needing physical replacement. However, this reprogrammability opens doors to several attacks such as Reverse Engineering, Code Modification, Fuzzing Attacks and Phlashing Attacks.	The attacks exploit security gaps to manipulate or damage the ECU software, leading to operational failures and the need for costly hardware replacements.	[38]
	Telematics Control Unit (TCU)	Tampering	Modifying TCU can manipulate the device's behaviour to benefit the attacker. By altering the transmission route's properties, communication attacks can occur on both physical and virtual levels.		[35]
		Denial of Service	Unauthorized persons launch DoS attack on TCU, can obstruct all incoming and outgoing communications of the vehicle. Since in-	The entire communication system is unavailable.	[35]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
			vehicle functions rely on networking protocols facilitated by the TCU		
		Elevation of privilege	The attacks can vary in nature, including erasing Diagnostic Trouble Codes (DTCs), altering settings, activating subscription-based features, facilitating further unauthorized access, or irreparably damaging the vehicle's electronics, as seen in ransomware attacks.	Enable attackers to modify data, harm connected devices, or damage the TCU itself.	[35]
		Information Disclosure	Attackers execute Man-in-the-Middle (MITM) attacks by intercepting wireless transmissions with a promiscuous antenna. A significant risk of such attacks is information leakage, primarily due to the combination of intercepted packets and weak data encryption.	Information leakage can lead to privacy issues, especially with the unauthorized gathering of sensitive data, including photos taken by a camera.	[35]
		Repudiation	The issue allows the attacker to either steal important information or send fake data to the vehicle's system without being traceable. If the vehicle's system accepts this fake data as real, it cannot be trusted anymore, and its functions might not work as they should.	The action could affect how well the vehicle operates and even put the driver's safety at risk, particularly in situations where the vehicle needs to automatically call for help e.g. after an accident.	[35]
	TPMS	Spoofing	TPMS lacks cryptographic mechanism, transmitting fixed sensor IDs in every packet, which could potentially allow vehicle tracking. Furthermore, transmitted messages lack authentication, and the ECU does not validate the received data. Consequently, an attacker could exploit this vulnerability by sending a malformed message to trigger a false tire pressure warning in another vehicle on the highway or even disable the TPMS ECU.	False tire pressure warnings cause unnecessary concern or actions from drivers, or in more severe cases, disable the TPMS ECU. Such disruptions not only compromise vehicle safety but also erode trust in the reliability of critical safety systems.	[24], [35]
	Onboard Diagnostic Port (OBD)		Once an external device establishes a connection to the OBD port, it gains the capacity to both send and receive data with the vehicle's ECUs. This unrestricted communication link could potentially be misused to deploy harmful payloads into the vehicle's network.	The potential repercussions of these security breaches are significant, including the possibility of remote control over vehicles. This underscores the critical necessity for implementing stringent security protocols within OBD port systems.	[6]
In- vehicle network (CAN, LIN, FlexRay and Ethernet protocols)	CAN	Spoofing CAN protocol	The primary issue with the CAN bus system is its absence of authentication and encryption, making it vulnerable to attacks. Intruders can manipulate vehicle functions by assigning CAN IDs and data fields.	An attacker can gain control over various automotive functions—such as stopping the engine, disabling the brakes, locking the wheels, and altering the display—by directly injecting fraudulent CAN messages through the OBD-II port [32].	[24], [32], [38]
		CAN and J1939 buses vulnerabilities	An unauthorized node has the capability to intercept and listen to any communications within the bus system. This vulnerability stems directly from the flaws in the arbitration mechanism. Consequently, an attacker can exploit this weakness to inject unauthorized frames into the network.	An attacker can exploit this weakness to inject unauthorized frames into the network.	[38]
		DoS in CAN	In an open bus system, transmitting a frame with a Controller Area Network (CAN) ID of 0 effectively halts all other traffic, as	This could interfere with the normal functioning of the vehicle, as ECUs are critical for controlling	[24], [32], [78], [79]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
			this ID is assigned the highest priority. Should a gateway be in place that filters and only permits specific IDs, it results in merely delaying the lower priority frames. Conversely, higher-priority frames proceed to be transmitted seamlessly. This design rationale partly explains why standard OBD-II diagnostic frames are assigned IDs starting from 0x7DF and above, positioning them at a lower priority level [78].	various aspects of the vehicle's operations, such as engine management, braking systems, and more.	
		Fuzzing in CAN	The attack can provoke a delayed response to certain conditions or serve as a denial-of-service by delaying message delivery. Its stealth lies in not raising error counters, with the sole indication being the tardy arrival of frames. Without prior timing analysis to determine maximum frame latencies, this attack mimics a transient timing fault, making it exceptionally hard to detect.	This could interfere with the normal functioning of the vehicle, as ECUs are critical for controlling various aspects of the vehicle's operations, such as engine management, braking systems, and more.	[24], [78]
		Bus-off in CAN	This attack targets the CAN transmission (TX) pin in a low-level protocol assault, focusing not on a single frame but on all frames emitted by a specific ECU. This strategy drives the Transmit Error Counter beyond 255, triggering the ECU's CAN controller to automatically enter a bus-off state, effectively disconnecting it from the network.	The target ECU is driven offline.	[32], [78]
	LIN	LIN Bus error-handling mechanism	The vulnerability in the LIN bus's error-handling mechanism can be exploited for attacks. Typically, when a collision is detected, the legitimate sender node halts packet transmission. This behaviour inadvertently provides an opening for malicious nodes to transmit a fraudulent message, effectively replacing the legitimate one.	The LIN bus is a cost-effective serial communication protocol designed to supplement the CAN bus. It is primarily utilized for controlling vehicle body functions, like seat adjustments and door operations. While the risk associated with LIN bus attacks may not be as high as those targeting the CAN bus, they still represent a significant security concern for high-speed vehicles.	[32]
	FlexRay	Read	The attack can be performed from any ECU in the FlexRay network. The absence of confidentiality measures on the FlexRay bus allows attackers to intercept and read all transmitted data, potentially relaying this information to a distant location through a wireless gateway.	FlexRay is designed to meet the future demands of automotive communication with its high data transfer rates, cost-efficiency, superior reliability, and adaptable data exchange capabilities. If secret keys, proprietary information, or private data are communicated over the FlexRay bus, attackers can effortlessly access and learn this sensitive information.	[32], [80]
		Spoofing FlexRay protocol	The attack can be performed from any ECU in the FlexRay network. An attacker is capable of fabricating and injecting messages. This vulnerability allows for the spoofing of messages that can target any Engine Control Unit (ECU), purporting to originate from any source.	An attacker can effortlessly craft and dispatch diagnostic messages over the FlexRay bus, prompting ECUs to execute arbitrary commands.	[32], [80]
		Close Proximity vulnerabilities	This threat arises from the use of short-range communication technologies, such as Bluetooth, tire pressure monitoring systems (TPMS), or keyless entry and ignition systems. It permits the execution of malicious code from any device previously paired	An attacker gains unauthorized access to a device or network, allowing them to eavesdrop on communications, steal sensitive information, or assume control over device functions.	[38]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
			via Bluetooth. An attacker can exploit a compromised device to target the vehicle's engine control units without the driver's knowledge. Despite the existence of secure Bluetooth protocols developed through cryptographic algorithms, their inefficiency often leads to their exclusion from commercial implementations. Moreover, these protocols do not safeguard against the injection of malicious code through memory exploitation.		
	Ethernet	ARP cache poisoning	An attacker can alter the ARP cache of a host by sending forged ARP requests or replies.	This can result in the attacker intercepting or modifying data meant for the legitimate host, leading to man-in-the-middle attacks.	[89]
		Denial of Service	The flashing process utilizes Diagnostics over IP (DoIP), and due to the large files being transferred, this communication operates over TCP. Tcpkill, a component of the dsniff toolkit designed for network assessments, monitors ongoing TCP connections on specified hosts. It disrupts these connections by injecting packets flagged with RST into the communication channel, causing the connection to be terminated. When combined with a DoS attack and spoofing, it can block a legitimate network host.	This lead to the communication is dropped.	[89], [90]
		SOME/IP Spoofing and Spoofing in general	An attacker sends forged packets that the receiving host cannot distinguish from the expected ones.	This can lead to unauthorized access or manipulation of vehicle functions, potentially causing disruptions in critical vehicle operations. Specifically, SOME/IP spoofing can allow the attacker to impersonate legitimate services or devices in the vehicle network, leading to further exploitation or disruption of the system.	[890], [90]
		TCP Hijacking	An attacker intercepts and takes over a TCP session by using spoofed packets to disrupt the sequence number flow, effectively replacing a legitimate host in the communication.	As a result, subsequent packets from the legitimate host are discarded due to outdated sequence numbers, effectively seizing control of the session and potentially leading to data theft or manipulation.	[89]
External network communication (Backend operations, Systems, Cloud Computing, Wi-fi and Cellular network)	Over-the-air (OTA) updates	Tampering and Malware injection by Privilege Accessing	- An attacker employs various methods to gain privileged access, including insider threats, such as disgruntled employees sharing admin credentials, social engineering tactics like phishing, and exploiting configuration weaknesses, such as default or blank passwords. This includes a broad range of attack techniques such as dictionary attacks, Pass-the-Hash, credential stuffing, and password spraying. - The cryptographic keys used for software or metadata signing were compromised, potentially through social engineering tactics.	- Uncover confidential details, reverse-engineer firmware, or analyse differences between firmware versions to pinpoint security enhancements and identify vulnerabilities that have been addressed. - Block the installation of updates, hindering the correction of software issues in vehicles. - Trigger the failure of one or more ECUs in the vehicle, resulting in the loss of access to the vehicle or specific functionalities. - Gain control over the vehicle's ECUs and potentially the entire vehicle.	[6], [81]

Category	Asset name	Threat	Threat Scenario	Consequences	Reference
	Remote access control the AVs by using external network	Tampering and Spoofing	The Tencent Keen Security Lab researchers exploited vulnerabilities to compromise the Tesla Autopilot Electronic Control Unit (APE) by remotely accessing it through external network connections like 3G/Wi-Fi. This allowed them to gain root access to the APE and manipulate the steering system from a remote mobile device, demonstrating a significant cybersecurity threat. They also disrupted the autowipers function using adversarial examples and misled the Tesla into the wrong lane with minor road modifications.	The driver lost control of the steering system, autowipes function and lane detection.	[91]
		Command Injection Attack, Network Eavesdropping, Tampering	The researchers exploited the vehicle's cellular connection (via the Uconnect system) which allowed the vehicle to communicate over Sprint's cellular network. They targeted this connection to gain initial remote access to the vehicle's systems from anywhere in the country without physical interaction.	The driver lost control of the vehicle.	[69]

Appendix B – Mitigation of Threat in AVs

Category	Asset Name	Specific Threats	Mitigation Strategies	Reference
Sensors	GPS	Spoofing and Tampering	- Implement sensor fusion to verify data from multiple sources. - Use Cryptographic techniques which are encrypting the GPS L1 P(Y) code and implementing navigation message authentication (NMA) to embed signatures in satellite signals, ensuring their authenticity.	[32]
	LiDAR	Spoofing and Denial of Service (DoS)	- Emitting laser pulses multiple times in one direction and narrowing the receiving angle to reduce the LiDAR's receiving time also decreases vulnerability by limiting the window for attacks. - Introducing randomness in the LiDAR's operation, like varying the rotation speed and the direction of laser emission, can also protect against attacks. - Employing redundancy through multiple LiDARs or multisensor fusion enhances the reliability of readings, though it may increase costs and complexity. - Side-channel authentication uses physically leaked information, such as power consumption or electromagnetic radiation, from a cryptographic device in a car performing AES calculations. The system only accepts echoes with this specific modulation, making it challenging for attackers to mimic. - To protect camera and LiDAR lenses, special coatings are applied to prevent scratches, dust, and glare while maintaining visibility in various weather conditions.	[32], [63], [92], [93]
	RADAR	Spoofing	The study [32] Recommend integrating sensor fusion techniques to guard against the introduction of counterfeit or altered sensor data, potentially averting erroneous judgments by the autonomous system.	[6], [32]
	Ultrasonic sensor	Spoofing	- Shifting the parameters of waveforms to authenticate physical signals, enhancing the security of sensor data. - Utilizing two or more sensors to detect attacks, thereby improving obstacle detection or localizing attackers.	[32]
	GNSS	Spoofing	Implement enhanced surveillance on the GNSS system using differential GNSS. This method corrects various inaccuracies within the GNSS system, thereby offering more precise positional data. The advantage of differential GNSS lies in its ability to pinpoint locations with high accuracy, which can be utilized to identify even minor discrepancies in position errors.	[63]
		Jamming	To ensure a backup source of location data in the event of GNSS failure, it is advisable to incorporate an onboard Inertial Navigation System (INS) or utilize dead-reckoning methods based on simple accelerometers, like those in contemporary smartphones. This approach would allow the ITS-S to ascertain its position using entirely internal data, with only occasional and brief references to GNSS for adjustments to waypoints.	[63]
	Camera	Tampering	- The optical characteristics and vulnerabilities of cameras make it challenging to design a completely secure camera system. However, employing strategies such as redundancy, using removable near-infrared-cut filters, and incorporating photochromic lenses can offer a level of protection from various angles. While these countermeasures are beneficial, they also come with their own set of limitations or may introduce new challenges to the system's overall security. - To protect camera and LiDAR lenses, special coatings are applied to prevent scratches, dust, and glare while maintaining visibility in various weather conditions. Seals around the camera housing keep out water and contaminants, ensuring functionality in different environments. Physical locks or secure mounting systems also guard against unauthorized removal or tampering, enhancing overall security.	[32], [92],[93]

Category	Asset Name	Specific Threats	Mitigation Strategies	Reference
In-vehicle Systems	ECUs	Tampering and external manipulation	- Encryption can secure communications between the Electronic Control Unit (ECU) and the system's other components by encrypting them, making unauthorized data interception or manipulation challenging. - Integrating intrusion detection systems can help identify and counteract tampering efforts. - Implement a secure key management process. - Use dedicated and tamper resistant hardware security modules (HSMs) to prevent errors common in software implementations and ensure more efficient use of resources. It is important to verify the quality of hardware cryptography, as any flaws will affect all associated software. - Coatings on ECU hardware serves to protect components from physical manipulation and environmental damage, enhancing overall durability. - HSMs incorporate physical security measures like tamper-evident seals, intrusion detection systems, and hardened enclosures. These features protect against tampering and environmental risks, ensuring the integrity of the HSM and the security of cryptographic keys from unauthorized access.	[35],[63], [96], [97], [98]
		Repudiation	- Using digital signatures with asymmetric keys from trusted third parties is a highly effective method for ensuring non-repudiation. This approach significantly enhances the recipient's confidence by confirming that the message truly originates from the claimed source.	[35]
	ECU port	Tampering	- Utilizing firewalls to segment the network and limit direct access to the ECU port. - Computing and storing hashes of sensor data by the ECU to verify against received data for tampering signs. - The ACN system digitally signs the data sent to the ECU, ensuring data authenticity and integrity. - Implementing strong authentication to ensure that only authorized personnel or systems can access the ECU port, preventing unauthorized tampering.	[35]
	TCU	Tampering	- The Secure On-board Communication (SecOC) protocol can be deployed to create secure communication channels between the ECU and other components, like TCU. Additionally, the SecOC protocol aids in secure bootstrapping during the initialization phase, ensuring the ECU is authenticated before starting data exchanges. - A Trusted Platform Module (TPM) can enhance the speed of encryption and decryption processes through hardware acceleration.	[35]
In-Vehicle Network	CAN	Impersonation, Replay and Denial of Service	- LiBrA-CAN: A secure broadcast authentication protocol using symmetric primitives, addressing computational costs, bus load, and execution time, but not suited for pre-built CAN buses. - TrustNet: A technique for using virtual and unreliable channels over CAN, offering protection with minimal overhead and legacy system compatibility. - CAN Arbitration and Security Features: Utilizing CAN's inherent arbitration, error detection, and fault confinement mechanisms to invalidate malicious packets. - Secure Communication Architecture: A scheme for secure ECU communication over different channels with a gateway unit, proposing the AEGIS encryption scheme for improved performance. - CAN ID Shuffling: A moving target defense method to hinder attackers from identifying CAN IDs, data frame cycles, and data field formats, implementable without altering existing CAN standards.	[24]
		Spoofing, Denial of Service (DoS)	Intrusion Detection system for CAN - CopyCAN intrusion detection method involves monitoring the CAN network to identify if a node has been disconnected by analyzing the ECU's error counter. - Combining a firewall and hardware-based mechanisms. It features a packet filtering system tailored to security protocols and CAN flexibility, alongside DoS defense and access controls. The innovative firewall utilizes a microcontroller and a hardware security module for encryption, with potential integration into the vehicle's central gateway.	[24]

Category	Asset Name	Specific Threats	Mitigation Strategies	Reference
	Ethernet	ARP cache poisoning	Turn on static ARP cache	[89]
		Denial of Service	Implement filters and firewalls to defend against DoS attacks.	[89]
		SOME/IP Spoofing and Spoofing in general	- Implement rigorous authentication mechanisms. This ensures that only legitimate entities can send and receive messages, preventing unauthorized access and manipulation by spoofed packets. - To enhance security, consider using protocols like Security Onboard Communication (SecOC) and Transport Layer Security (TLS). Additionally, the Security Real-time Transport Protocol (SRTP) provides encryption, integrity, and message authentication. This can be effectively combined with Multimedia Internet KEYing (MIKEY), a key management solution that supplies session keys for SRTP.	[89]
		TCP Hijacking	Use encryption to protect sequence numbers from being intercepted.	[89]
External Network communication	V2X		Blockchain technology can offer a secure method for vehicle-to-infrastructure communications, safeguarding the data's integrity and confidentiality. Additionally, it facilitates a reliable, decentralized framework for both vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) exchanges.	[6]
	Software	Flashing	Secure boot mechanisms are crucial for ensuring the trustworthiness—authenticity and integrity—of software upon execution. These mechanisms typically involve the manufacturer signing the software using a private key. During the boot process, the software's signature is checked against a corresponding certificate. If this verification fails due to any alteration in the software, the system should respond appropriately, such as by initiating an emergency program to maintain the vehicle's functional safety.	[63]
		Reverse engineering and Tampering of their binary code	Code obfuscation methods can be employed to hinder the reverse engineering of mobile applications for smart cars. To facilitate the identification of alterations in the mobile application binary code, the applications should be digitally signed. Additionally, root detection systems can be utilized to ascertain whether the device has been rooted.	[63]
	OTA firmware updates	Tampering and malware injection or Hijacking	- Provide robust authentication methods, such as digital signatures, to verify the authenticity and integrity of firmware and prevent the installation of malicious firmware and the dissemination of malware. It is also important to prevent rollback to vulnerable firmware versions. Additionally, using encryption can help shield against the analysis of binaries, thus protecting intellectual property and preventing the exposure of zero-day vulnerabilities. - Implement secure boot mechanisms by signing software with a manufacturer's private key and verifying this during boot-up. If the software is altered and verification fails, the system should trigger a response, such as running an emergency program, to maintain vehicle safety.	[6], [63]
		Malfunctioning vehicles	Consider implementing access controls for OEM back-end servers and establishing recovery protocols for errors, such as reverting to a previous system state if the OTA (Over-the-Air) update fails to boot successfully.	[63]
	Bluetooth, TPMS and Key fob or Keyless entry	Close Proximity Vulnerabilities	Encryption and cryptographic checksum.	[38]
	All messages	Replay attack	Timestamp all messages	[63]
	Backend Office	Tampering, Non-repudiation	Ensure that audit logs are securely maintained. It is essential to record security events and keep track of who accesses the logs, safeguarding them against unauthorized exposure. Additionally, logs are required for integrating devices.	[63]

Category	Asset Name	Specific Threats	Mitigation Strategies	Reference
	Radio Frequency	Hijacking, Physical attack, DDos, Jamming	Manage radio spectrum frequencies and implement frequency agility.	[63]
	The vehicle's Uconnect system	Command Injection Attack, Network Eavesdropping, Tampering	- Implementing a robust and secure update mechanism for vehicle firmware and software was highlighted. This includes secure delivery and verification of updates to prevent malicious modifications. - The need to secure network services, especially those like D-Bus, which should not be exposed over the network. Services should be properly authenticated and limited to local access unless necessary for remote functionality. - Strengthening the encryption on the Wi-Fi hotspot and ensuring strong, unpredictable passwords are generated. The use of WPA2 should be standard, and steps should be taken to avoid predictable password generation based on time.	[69]

Appendix C – TARA Methodology

	Inputs	Requirements and Recommendations	Outcome
Asset Identification	Item Definition (15.3.1.1): Item's definition, which outlines its boundaries, functions, and preliminary architecture. Supporting Information (15.3.1.2): Additional details from cybersecurity specifications provide insight into the cybersecurity requirements or features related to the item.	Damage Scenarios (RQ-15-01): Scenarios where damage could occur, encompassing adverse consequences related to the item's functionality, harm to road users, or relevant assets. Assets Identification (RQ-15-02): Identifying assets with cybersecurity properties that, if compromised, could lead to damage scenarios.	Damage Scenarios (WP-15-01): Document lists and describes the identified damage scenarios, providing a comprehensive understanding of potential adverse consequences. Assets with Cybersecurity Properties (WP-15-02): Document describes the identified assets along with their associated cybersecurity properties, facilitating a clear understanding of assets susceptible to compromise.
Threat Scenario Identification	Prerequisites (15.4.1.1): Item Definition Supporting Information (15.4.1.2): Additional details from cybersecurity specifications, damage scenarios, assets with cybersecurity properties.	Identification of Threat Scenarios (RQ-15-03): Threat scenarios must be identified, including the targeted asset, compromised cybersecurity property of the asset, and cause of compromise of the cybersecurity property. Methods for Identifying Threat Scenarios: Elicitation of malicious use cases or Threat modelling approaches (EVITA, TVRA, PASTA, or STRIDE)	Threat Scenarios (WP-15-03): A document listing and describing identified threat scenarios, including details about targeted assets, compromised cybersecurity properties, and causes of compromise for each scenario.
Impact Rating	Prerequisites (15.5.1.1): Damage scenarios Supporting Information (15.5.1.2): Item definition and assets with cybersecurity properties.	Assessment of Damage Scenarios (RQ-15-04): Evaluated for potential adverse consequences in impact categories related to (S) safety (F) financial (O) operational (P) privacy Note 1: The document doesn't provide predefined relationships between different impact categories, allowing organisations flexibility in prioritizing them based on their specific contexts.	Impact Ratings with Associated Impact Categories (WP-15-04): The output of the impact rating process, containing impact ratings assigned to each damage scenario along with their associated impact categories.

	Inputs	Requirements and Recommendations	Outcome
		Note 2: Additional impact categories beyond the specified four can be considered as deemed necessary by organisations. Note 3: Rationale and explanation for additional impact categories should be shared within the supply chain according to Clause 7 of the ISO document. Impact Rating Determination (RQ-15-05 and RQ-15-06): Impact ratings for each damage scenario must be categorized as severe, major, moderate, or negligible, with specific guidance provided for financial, operational, and privacy-related impacts. Safety-related impact ratings should be derived from ISO 26262-3:2018.	
Attack Path Analysis	Prerequisites (15.6.1.1): Item definition or cybersecurity specifications, documented threat scenarios. Supporting Information (15.6.1.2): Additional helpful information includes weaknesses from cybersecurity events, weaknesses found during product development, architectural design details, previously identified attack paths (if available), and vulnerability analysis results.	Analysis of Threat Scenarios (RQ-15-08): Threat scenarios must be analysed to identify potential attack paths that adversaries could exploit to realize those threats. Note 1: Attack path analysis can be based on top-down or bottom-up approaches, Note 2: partial attack paths that do not lead to the realization of a threat scenario may be stopped. Association with Threat Scenarios (RQ-15-09): Each identified attack path must be clearly associated with the specific threat scenarios it enables. Note 3: Attack paths may be incomplete or imprecise in early stages of product development, but they can be updated as more information becomes available, such as through vulnerability analysis.	Attack Paths (WP-15-05): The output of the analysis, consisting of identified potential pathways or sequences of events that attackers could exploit to compromise the system's security and cause significant impacts.
Attack Feasibility Rating	Prerequisites (15.7.1.1): Attack Paths Supporting Information (15.7.1.2): Architectural Design, Vulnerability Analysis	Determination of Attack Feasibility Rating (RQ-15-10): Attack feasibility ratings are determined for each attack path as High, Medium, Low, or Very Low based on the effort required for the attack. Additional Guidelines (RC-15-11): The method can be chosen. Attack potential-based. CVSS-based. Attack vector based.	Attack Feasibility Ratings (WP-15-06): This document records the assessed attack feasibility for each attack path, aiding decision-making and risk management in cybersecurity activities related to system development.
Risk Value Determination	Prerequisites (15.8.1.1): Threat Scenarios, Impact Ratings with Associated Impact Categories, and Attack Feasibility Ratings.	Determination of Risk Value (RQ-15-15): The risk value for each threat scenario should be calculated based on the severity of impact and the feasibility of attacks. Note 1: If a threat scenario corresponds to multiple damage scenarios or impacts multiple impact categories, separate risk values can be determined for each. Note 2: If a threat scenario corresponds to multiple attack paths, the associated attack feasibility ratings can be aggregated, such as by assigning the maximum rating among the corresponding attack paths.	Risk Values (WP-15-07): This document presents the determined risk values for each threat scenario, providing a clear assessment of the level of risk associated with each scenario.

	Inputs	Requirements and Recommendations	Outcome
		Range of Risk Values (RQ-15-16): The risk value of a threat scenario should fall between (and including) 1 and 5, where 1 represents minimal risk.	
Risk Treatment Decision	Prerequisites (15.9.1): Item Definition, Threat Scenarios, Risk Values, Cybersecurity Specifications, Impact Ratings, Attack Paths, and Attack Feasibility ratings. Supporting Information (15.9.1.2): Previous risk treatment decisions for similar components.	Determining Risk Treatment Options (RQ-15-17): For each threat scenario, one or more risk treatment options must be determined based on its risk value: a) Avoiding the Risk b) Reducing the Risk c) Sharing the Risk d) Retaining the Risk Note: Reasons for choosing to retain or share the risk should be documented as cybersecurity claims and are subject to cybersecurity monitoring and vulnerability management.	Risk Treatment Decisions (WP-15-08)